GAS PRICES — From the Ground and Before, to the Pump and Beyond

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The complete anatomy of what you pay at the pump, who touches it along the way, what breaks it, who profits, and where it goes next — assembled from a forensic, non-biased research archive spanning geology, markets, wars, court dockets, and the 2026 oil crisis.

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HOW TO READ THIS DOCUMENT

Every number, name, and event in this document comes from a dated research archive compiled section by section over sixteen research dives, with each source graded by authority: official statistics and court records (highest), major financial media (high), industry and advocacy material (flagged as such), and analyst estimates (flagged as estimates). Where the evidence conflicts, both sides are presented. Where allegations were made, the outcome — conviction, acquittal, dismissal, or unresolved — is stated. Nothing here is invented. Where the archive could not verify a claim, the claim is marked as such rather than presented as fact.

A note on the one question everyone asks first: *who is to blame for what you pay?* This document does not answer that question with a name. It answers it with a map — of molecules, money, machines, laws, and people — so that you can see every hand that touches the price, and decide for yourself.

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PART I — BEFORE THE GROUND

Chapter 1: The Inheritance

Long before there was a gas station, an oil company, or a price sign, there was a shallow sea, and in it, a rain of tiny dying things.

Petroleum begins with plankton and algae — the diatoms, the floating drifters of ancient oceans. When they died, a tiny fraction of that organic matter, roughly one-tenth of one percent, escaped the ordinary fate of being eaten and rotted away. It settled into mud on the sea floor, in still, oxygen-starved water, and was buried. Layer by layer, sediment pressed down. Heat and pressure — the slow patience of geology — cooked that buried organic sludge through stages geologists name with precision. First comes diagenesis, where the material becomes a waxy substance called kerogen. Then comes catagenesis, the "oil window," roughly two to four kilometers deep and between about 60°C and 120°C, where kerogen cracks into the liquid we call crude. Then, deeper still, the "gas window," where heat breaks the liquid into natural gas. And finally, at depths where no hydrocarbon survives, the zone where heat destroys what it once created.

The numbers that govern this process have a strange intimacy. It takes, on average, tens of millions of years — around sixty million by some estimates — for sediment to bury organic matter deep enough to cook. The pressure rises about twenty-five atmospheres for every hundred meters of depth. The whole staircase is measured in geologic time, while the withdrawal of the finished product is measured in seconds.

Here is the first hard truth of gasoline economics, and it is so obvious it hides in plain sight: **oil is a stock, not a flow.** Nature manufactured a fixed, one-time inheritance of concentrated ancient sunlight, and every barrel we pump is subtracted from it permanently. "A finite amount of these hydrocarbons is present within the earth," as one geology reference flatly notes. "They cannot be manufactured when known reserves are depleted." The price of gasoline is, at bottom, the price of rationing an inheritance — which is why every part of this document, from cartels to wars to windfall taxes, is really an argument about how to ration that inheritance.

One myth to clear away at the start: oil does not come from dinosaurs. The marketing image of the brontosaurus is wrong; your gasoline is mostly ancient sea plankton, filtered through geology.

And one more foundation for what follows: not every rock that holds oil gives it up easily. Most oil sits trapped in fine-grained source rock — shale — from which it must either migrate upward into porous reservoir rock capped by an impermeable seal (the classic "trap" that geologists hunt), or be forced out by technology. Rock that has never finished cooking, called oil shale, can be heated artificially to simulate geology — a patent for extracting oil from stone was granted, remarkably, in 1694. Rock that has cooked but never migrated is what the shale revolution fracks. From the very beginning, then, the cost of a gallon has depended on *which kind of rock* it came from — the cheapest-to-tap inheritance long gone, the harder layers defining the price floor of the modern world.

Chapter 2: Getting It Out — What a Barrel Costs Before It Is a Barrel

To understand pump prices, you must understand that the price of crude oil is built, from the bottom up, out of physical extraction costs — and those costs vary wildly depending on where and how you drill.

Picture the global oil supply as a staircase. On the top step sit the cheapest barrels on Earth: the giant, pressurized fields of the Middle East, where Saudi Arabia can lift a barrel for a handful of dollars. On the middle steps sit Canadian oil sands and deepwater Brazil. On the bottom steps, the most expensive: new shale wells in America.

Here is the modern cost map, assembled from the archive's most authoritative sources:

Saudi Arabia and the Gulf can produce for roughly $10–$15 per barrel, with fiscal breakeven prices (the oil price their governments need to balance their budgets) far higher — Saudi Arabia's fiscal breakeven sits around $86–$92 by International Monetary Fund estimates, the United Arab Emirates' near $45–$50, the lowest in OPEC. This gap between lifting cost and fiscal breakeven is one of the most important — and least understood — numbers in world politics: a country can make money on every barrel and still go broke as a nation.

Canada's oil sands need roughly $40–$55 operating costs, and new growth projects assume around $55.

American shale is the world's price-setter at the margin, and its costs are the hinge on which global prices swing. A new shale

well in the best zones of the Permian Basin breaks even around $60–$70 a barrel; the average new well across America needs about $70. But shale carries a hidden treadmill: a typical well loses about three-quarters of its production in its first year, which means the industry must constantly re-drill simply to stand still. Maintaining America's 13-million-barrel-a-day output requires replacing roughly 430,000 barrels per day of *lost* production every single month. As the sweetest drilling spots deplete, analysts project the average new-well breakeven climbing toward $82–$85 by 2030 and $95 by the mid-2030s. America has already drilled much of its cheapest inventory — some analysts estimate only about six years of top-tier well locations remain at current pace.

The other engines. The rest of the world's new supply runs on three engines. Brazil's pre-salt fields set national production records through 2026 — over 4.2 million barrels a day, with Petrobras running its refineries above nameplate utilization. Guyana, from a standing start in 2019, passed 900,000 barrels a day and crosses a million in 2027, becoming South America's second-largest producer. And Kazakhstan's Tengiz megaproject plateaued near a million barrels a day — but 2026 taught its cautionary lesson: more than 80 percent of Kazakh exports ride a single pipeline, the Caspian Pipeline Consortium across Russia, whose repeated wartime disruptions forced the country, unprecedentedly, to cut its production target on transit-security grounds. Supply, in the end, is not just geology. It is geology multiplied by transit.

The shale lesson that rewrote market physics: shale is short-cycle supply. A shale well can be drilled, fracked, and producing within months, and it can be left unfinished (a "drilled but uncompleted" well, or DUC) until prices justify finishing it. Old-fashioned mega-projects take a decade. This asymmetry — fast shale versus slow everything-else — explains why every attempt by OPEC to hold prices high eventually summons American supply to cap them, and why the world discovered in 2026 that the mechanism works in reverse too: at $50 oil, shale retreats almost as quickly as it arrived.

One 2026 voice captured the squeeze from the inside. A Texas oil executive told the Dallas Fed survey: "The administration is pushing for $40 per barrel crude oil... with tariffs on foreign tubular goods, [input] prices are up, and drilling is going to disappear." That sentence contains the whole modern price range: cheap enough to please consumers, high enough to keep the marginal barrel flowing — and the corridor between them is narrow and getting narrower as the easy rock runs out.

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PART II — WHO OWNS THE INHERITANCE

Chapter 3: From Standard Oil to the Seven Sisters to the State

Follow the ownership of oil across 160 years and a pattern emerges: whoever holds the *spare capacity* holds the price — and every regime of price stability in history was engineered by someone sitting on a cushion of idle barrels.

Act One: the monopolist (1870–1911). John D. Rockefeller founded Standard Oil of Ohio in 1870, and within a decade his company controlled roughly 90 percent of American refining — achieved partly through the infamous South Improvement Company scheme of 1871–72, a secret cartel with the railroads that would have crushed independent producers had it not leaked, sparking the Oil Creek uprising that killed it. Standard Oil survived the scandal; its competitors largely did not. In 1882 the trust formalized the empire; in 1911, in a unanimous Supreme Court decision under the Sherman Antitrust Act, the trust was broken into 34 successor companies. The breakup's most famous irony: Rockefeller's wealth approximately doubled, because the successors' combined value eventually exceeded the whole. Every one of today's oil majors descends from that 1911 split: Standard of New Jersey became Exxon; Standard of California became Chevron; Standard of Indiana became Amoco, later swallowed by BP; Ohio Oil became Marathon; the original Standard of California partnership built what would become Saudi Aramco.

Act Two: the Sisters and the posted price (1911–1973). The successor companies plus two Europeans — Royal Dutch Shell and Anglo-Persian (BP) — became the "Seven Sisters," which by 1950 controlled about 85 percent of world reserves and simply *posted* the price of oil, as a store posts the price of milk. For three decades the world enjoyed the most stable energy prices in its history: roughly $1.80–$2.50 a barrel, equivalent to $12–$15 in today's money. This stability was not natural. It was engineered — by the majors, and, increasingly, by an American regulator few remember.

Act Three: the Texas Railroad Commission — the original OPEC. Here is one of the great forgotten stories in economic history. American oil law operated on the "rule of capture": whoever pumped first owned the oil, regardless of who owned the land above — which made over-drilling a competitive necessity and prices a yo-yo. When the giant East Texas field flooded a Depression market in 1930–31, local crude fell toward ten cents a barrel. The Texas Railroad Commission — created in 1891 to regulate railroads — asserted the power to prorate: to order each well to produce only a fraction of its capacity, "for conservation." The courts struck its orders down as price-fixing; Governor Ross Sterling declared martial law in the oil counties in August 1931 and sent a thousand National Guard troops to shut the wells; "hot oil" smugglers ran the blockade anyway. But under commissioner Ernest Thompson — a force of personality who held the chair from 1932 to 1965 — the system held. By controlling roughly a third to two-fifths of American output, the Commission became the *de facto world swing producer*: it set allowables, and the world price

followed. When OPEC's founders sat down in Baghdad in 1960, they studied the Railroad Commission's records. OPEC was explicitly modeled on a Texas regulator.

Act Four: the state's revenge (1970–present). The American stabilizer expired of its own success: U.S. production peaked in 1970, and when the 1973 embargo hit, Texas had no spare capacity left to release. Into the vacuum stepped the producer states. The 1970s nationalization wave — Libya, Iraq, Saudi Arabia, Iran — transferred the Seven Sisters' concessions to national oil companies, and with them the pricing power. Today the map has inverted: national oil companies hold over 90 percent of world reserves, the descendants of the Sisters are diversified giants, and the price of your gasoline depends on the budget mathematics of Riyadh, Abu Dhabi, and Moscow as much as on any market.

The 2026 epilogue: in May 2026 the United Arab Emirates — holding a quarter of OPEC's spare capacity — left the organization entirely, the first major Gulf producer ever to do so, after years of quota disputes and an output collapse that made the exit momentarily painless. Analysts called it "the beginning of the end of OPEC" and "the end of the beginning" in the same news cycle. Whether it proves either is a question for Part XV. What matters for now is the through-line: **every stable price era was someone's deliberate act; every price collapse was someone's deliberate act ending.**

Chapter 4: The Counter-Example — What Happens When Nobody Is Allowed to Set the Price

Before we follow oil's price into the modern market, it is worth pausing on the great natural experiment that proves the rule: the United States, 1971–1981.

When Richard Nixon froze wages and prices in August 1971, oil was left under the controls longest. When the 1973 embargo tripled the world price, Washington's response was not to let the price rise but to legislate scarcity away: the Emergency Petroleum Allocation Act created a two-tier price ceiling ("old oil" capped at $5.25 a barrel, new oil higher) and pro-rata *allocation* — if supply fell to 90 percent, every buyer received 90 percent of their 1972 allotment, "fair and equitable" exceptions permitting.

The result was not stability. It was queues. The gasoline ceiling of 57 cents sat against a market-clearing price of roughly $1.50 — and a price that is not allowed to rise does not make the shortage disappear; it makes the shortage ration by other means: by line, by favor, by license plate. America got odd-even rationing, flag systems, stations running dry, "favored customer" arrangements, and hidden service charges that functioned as a black market wearing a white collar. Tier ceilings multiplied — 3 tiers by 1976, 5 by 1977, 8 to 11 by 1979 — breeding paper markets in "old oil entitlements" that redistributed rents among refiners without producing a single extra barrel. Controls were finally phased out beginning in 1980, capped by a 70 percent Windfall Profit Tax that raised barely a fifth of what it was designed to raise, and

killed outright by Ronald Reagan in January 1981. The most instructive sentence in the whole episode belongs to economists Kevin Hoover and David Henderson: the oil shock itself cost the U.S. about 1.4 percent of GNP — and the controls made the losses *larger than the shock*.

Hold this chapter in mind. Every modern government that "does something" about gasoline prices — China's bandwidth rule, Indonesia's frozen pump prices, Australia's excise holiday, America's strategic reserve — is an attempt to smooth the price signal without repeating 1974. Some succeed. Some fail in ways this document will map.

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PART III — HOW OIL GETS ITS PRICE

Chapter 5: The Benchmarks — Three Barrels and the Fiction That Prices Them All

There is no such thing as *the* price of oil. There are prices — plural — and the map of which barrel you mean is a map of power.

For most of living memory, crude was priced by whoever controlled it: the majors posted it (pre-1973), then OPEC posted it (1973–1986). After OPEC's pricing system collapsed in 1986, pricing migrated to two futures exchanges, and the world reorganized itself around two (now three) reference barrels.

Brent is a basket of North Sea crudes — today effectively "Brent plus WTI Midland," since American barrels were admitted to the basket in 2023 as North Sea production dwindled. Brent prices roughly 60 percent of the world's traded oil and is the anchor of the international market.

WTI (West Texas Intermediate) is the American barrel, physically delivered at a small Oklahoma town named Cushing, and it is the anchor of the futures market in New York.

Dubai/Oman anchors the Asian market, where the giant buyers of the Gulf's crude live.

Each benchmark is itself an *average of judgments*, not a single trade. Platts, Argus, and OPIS — the price-reporting agencies — collect bids, offers, and deals from traders and publish assessments; even Brent exists in several versions simultaneously (futures price, index, dated, cash). In normal times the versions

hug each other. In crisis times they can fly apart — in April 2026, with the Strait of Hormuz closed, "dated" physical Brent traded more than $25 above the futures price, the largest gap in a generation, because whoever needed a barrel *now* had to pay a war premium that a paper contract for next month did not carry. The price you pay at a gas station ultimately descends from these assessments, compiled in real time by human editors making judgment calls under conditions the rulebooks call "illiquidity." Keep that in mind when someone says "the market decided."

The Brent–WTI spread is a story about pipes. Until 2011, the higher-quality American barrel traded at a slight premium. Then the shale boom outran the pipelines: crude backed up at Cushing, and WTI fell to a $27-a-barrel discount at its extreme in 2011–12 — the same molecule, two prices, separated by a bottleneck. Building pipelines and lifting the export ban in 2015 re-anchored WTI to the world. Seven years later, in 2026, the spread inverted again for the opposite reason: America was insulated from the Hormuz crisis, and the home-market barrel became the world's *strongest*, at times trading *above* Brent — a map of geography drawn in dollars.

And the dollar itself? The "petrodollar" is real but routinely exaggerated: there was never a formal pact that expires, only a post-1971 architecture in which every major benchmark is priced in U.S. dollars, oil revenue recycles into American Treasury bonds, and both sides benefit. The dollar and oil moved in opposite directions for decades — a stronger dollar, cheaper oil — until America's shale boom made the U.S. an exporter and flipped the sign of the correlation in the mid-2010s, a regime economists

still debate. Roughly a fifth of world oil trade now settles in yuan, euros, and rupees — mostly by sanctioned sellers — without yet denting the dollar's dominance. The dollar layer is real; the conspiracy version of it is not in the evidence.

Chapter 6: The Casino on Top of the Reservoir — Futures, Speculation, and the Honest State of the Evidence

Every few years, when gasoline spikes, a politician points at the futures market and says *that* is the culprit — the speculators. The evidence, assembled over two decades of academic war, deserves a straight answer.

How it works, in one paragraph: a futures contract is a bet on next month's price, used by airlines, refiners, and farmers to lock costs in advance. Every bet needs a counterparty — someone to take the other side. "Speculators" are those counterparties: hedge funds, index funds, pension money. When they pile in on one side, the futures curve bends; when physical supply is tight, the curve bends into "backwardation" (spot above futures); when storage is full, it bends into "contango" (futures above spot).

The 2008 war, with its scorecard. After crude ran to $147 and gasoline to over $4, Congress held hearings. A Senate investigations report blamed index funds. The Commodity Futures Trading Commission's own study — surveying the academic literature — found the opposite: the dominant peer-reviewed studies (Büyüksahin & Harris, Brunetti, Gilbert, OECD) could find no evidence that speculator positions *led* prices; if anything, prices led positions. The Bank of Canada concluded speculation "does not seem to have played an important role" in the 2003–08 run-up. The one structural study allowing a speculative component (Juvenal & Petrella) still attributed only about 13 percent of price variance to it, versus 45 percent for global

demand shocks. And a scale check that should be memorized: at the 2008 peak, index funds' crude exposure equaled less than one day's world demand — 0.26 percent of a year's consumption. The honest synthesis: fundamentals dominate; a measurable but secondary speculative component existed; the "speculators did it" theory is not supported by the best evidence, and this document does not adopt it.

And then 2026 provided a live test. In the biggest oil spike since records began — caused by a war, as Part XV details — the positioning data showed managed-money traders *net short* crude oil at the very April price peak, betting on the fall while the physical market screamed. The gasoline rally that followed happened while speculators were *selling*. If a casino caused the fire, this was a casino full of people running the wrong way.

What *is* in the court record: manipulation of specific, narrow market moments by specific firms — squeezes at the Cushing delivery point, closing-price games, benchmark windows. The CFTC has collected over a billion dollars in such penalties, including the largest commodity-manipulation fine in history. That is a different claim than "the whole price is fake," and the distinction matters.

Chapter 7: Minus $37 a Barrel — The Day the Machine Broke

On April 20, 2020, the price of the world's most important commodity went *negative*: the May WTI contract settled at minus $37.63 a barrel, with panicked traders paying to get rid of oil.

Understanding why requires understanding the machine under the price. WTI futures do not settle in cash — they settle in *physical delivery* of 1,000 barrels per contract at Cushing, Oklahoma. Cushing exists as a delivery point because in 1983 it sat at the crossroads of America's early pipelines; today it is a tank farm of roughly 76 million barrels of working capacity amid thirty pipelines. A trader holding a futures contract into expiry must either sell it or *take delivery* — and taking delivery means you need somewhere to put the oil.

April 2020 was the perfect convergence: COVID lockdowns erased 16 million barrels a day of demand (six times the 2008 collapse); an OPEC+ price war flooded the market; and Cushing was filling toward its ceiling. Storage is never "full" in the literal sense — but space that is not pre-booked, accessible, and matched to pipeline schedules might as well not exist. By contract expiry, effectively all of it was spoken for. Traders who had bought paper oil as a punt suddenly faced physical delivery they could not accept, and with hours to go, they paid counterparties to take the contracts off their hands. Twenty-one minutes near the close destroyed the price.

Brent — a cash-settled basket with flexible North Sea logistics — never went negative. The lesson is architectural: **a benchmark**

is only as robust as its delivery point, and local conditions at one Oklahoma tank farm can hijack the price of a global commodity.

And the epilogue proves the machine was the problem: the very next day, the June contract settled at plus $10 — the futures market, denounced as broken, had corrected itself within twenty-four hours.

Now hold that architecture in mind and fast-forward six years. In July 2026, Cushing hit the *opposite* wall: with U.S. refineries running flat-out and exports gushing, storage fell toward 18–21 million barrels — the "tank bottoms," the operational minimum below which pumps and infrastructure cannot function. The market's cushion at the delivery point had evaporated, and WTI swung to *premiums* over Brent, futures backwardated $7 across three months — a "physical squeeze," the 2020 failure inverted. Same hub, opposite failure, seven years apart. The takeaway for the rest of this document: when you read about "oil prices," you are reading about a machine with hard physical limits at both ends — and 2026 pressed both.

Chapter 8: The Reserve — America's Underground Bank

If Cushing is the machine's valve, the Strategic Petroleum Reserve is the nation's savings account — and its 2021–2026 history is a masterclass in what a reserve can and cannot do.

Created by Congress in 1975 in direct response to the embargo, the SPR was designed to hold the *medium and heavy sour* crude that Gulf Coast refineries were built to process, stored in man-made salt caverns along the Texas and Louisiana coast (salt is self-sealing and cheap; the caverns were solution-mined into existence). It filled for three decades, peaking at 727 million barrels in 2009.

Then came the era of the drawdown. In November 2021 and March 2022, with gasoline surging after Russia's invasion of Ukraine, the Biden administration sold 180 million barrels — a million a day for six months, the largest release in history — cutting the reserve from about 600 million barrels to 347 million. It bought back some, slowly, at lower prices.

Then came the 2026 war. In March, as the largest supply disruption in the history of the oil market unfolded (Part XV), the Trump administration authorized a 172-million-barrel release over 120 days — faster than Biden's — as part of a 32-country International Energy Agency action totaling 400 million barrels, the largest collective release ever conducted, with Japan contributing 80 million, Germany 20 million, and dozens more. By August the SPR had fallen to roughly 285 million barrels, its lowest since 1983 — about 35 million barrels above the

operational floor of its infrastructure, with a government auditor warning that over a quarter of the remaining inventory could not be rapidly deployed because 1970s-era equipment was failing.

The accounting matters less than the physics. A 14-million-barrel-a-day supply shock can be cushioned by releases of 2.5–3.3 million barrels a day — for *weeks*. As one analyst put it during the crisis: high stocks buy time; they do not replace flow. And there is a second-order effect now building: refilling those tanks — America's promised 40 million barrels, Japan's 80 million, China's ongoing build — creates a floor of price-insensitive buying under the 2027 market, a "restock bid" that every forecaster must reckon with.

One last irony in the reserve's design: it was built for the oil America used to import (heavy, sour), while the shale era made America a producer of light, sweet crude the reserve cannot optimally hold. The bank's vault was designed for last century's deposits.

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PART IV — THE MACHINE IN THE MIDDLE: REFINING

Chapter 9: How a Barrel Becomes a Gallon — and Why That Costs What It Costs

Crude oil is useless to a car. The entire value of a refinery is conversion: taking a raw mixture of hydrocarbons and rearranging it into the molecules the world actually buys.

Picture the process as a kitchen with increasingly specialized stations. The first station is the simplest — the atmospheric distillation column, a giant still. It separates crude by boiling point into its natural fractions: the light naphtha that can become gasoline, the kerosene that becomes jet fuel, diesel, and the heavy residue at the bottom. A refinery with *only* this station is called a "topping" plant, and it makes relatively little gasoline per barrel.

The magic stations come next. The *fluid catalytic cracker* breaks heavy gas oil into high-octane gasoline — the workhorse of the gasoline age, invented in the run-up to the Second World War. The *hydrocracker* does similar chemistry with hydrogen, producing clean diesel and jet. The *catalytic reformer* converts mediocre naphtha into high-octane aromatics. The *alkylation unit* — the most expensive station per barrel of capacity — turns light olefins into the very highest-quality gasoline blendstock. Add a *coker* to grind residue down to nothing, and you have what the industry calls a "deep conversion" refinery.

Engineers quantify this kitchen with a single number: the **Nelson Complexity Index**, created by engineer Wilbur Nelson in

1960 as a pure cost metric — how expensive each unit is relative to the basic still. But cost bought *optionality*, and the index quietly became a power ranking. A complex refinery can run the cheap, heavy, sour crude nobody else wants — capturing discounts of $10–15 a barrel — and choose its product mix. America's refineries average a world-leading complexity of about 9.5, built over decades to feast on discounted heavy crude; the world's largest refinery, Reliance's Jamnagar complex in India, runs a complexity of 21. The spread between what refiners pay for crude and what they sell products for is called the "crack spread" — the refining industry's profit margin, and the second-biggest component of your pump price after the oil itself.

Chapter 10: The Disappearing Buffer — Why Refining Became the Bottleneck

Here is the structural story that explains more of your pump price in the 2020s than any other: the rich world's refining capacity stopped growing just as the world's demand stopped falling where expected.

America's last major new refinery was built in 1977. The count of U.S. refineries fell from 301 in 1982 to 130 by 2026 — though capacity fell much less than the count, because survivors grew larger and more complex. The closures accelerated: the pandemic killed marginal plants; California's stack of unique regulations made its refineries among the world's most expensive to run, and one by one they converted, shrank, or closed (Phillips 66's Los Angeles plant in 2025, Valero's Benicia in 2026), leaving the state with a refining cushion barely above its consumption and no pipeline to the rest of the country.

Europe went further, faster: roughly thirty refineries closed since 2009 (a 37 percent capacity cut), Scotland's Grangemouth converted to an import terminal in 2025, and the UK's nine refineries of 2000 became four. The drivers were environmental policy, carbon costs, and brutal competition from new Asian mega-refineries — 10 million barrels a day of capacity closed globally from 2019–2026 even as demand kept surprising to the upside.

The analytical institutions saw it coming and said so: the IEA warned that European refining was at risk, that new capacity additions were far below what a "no-transition" world would need,

and that the global product market would tighten — *if* demand held up. It held up. And then two wars struck the product chain directly.

Ukraine's drones versus Russia's refineries. Starting in 2024 and escalating violently through 2026, Ukraine converted cheap long-range drones into precision strikes on the one node of Russia's oil machine that could not be rerouted: its refineries. By mid-2026, Ukrainian statistics claimed more than half of Russian refining capacity disabled at various points; Russian crude runs fell to levels not seen in two decades; Moscow banned diesel exports outright — removing about a tenth of the world's seaborne diesel — and by autumn, Russian refineries were importing gasoline *from India*. A top-ten exporter of transport fuels became, structurally, an importer.

Nature joined the list: hurricanes strike the refining coast in the August–September peak season almost every cycle — Katrina and Rita in 2005 (a quarter of national capacity offline), Harvey in 2017 (the crude-falls-while-gasoline-soars signature), Ida in 2021 — with impact governed by exactly the variables of this chapter: refining exposure, inventory cushions, and season.

The Hormuz shock (Part XV in full) then removed a second block of product supply — the Gulf's own mega-refineries and its exports — at the very moment the system had no spare capacity anywhere else.

The result, in March–September 2026, was the most extreme refining-margin event in the data: American refining margins roughly doubled year-over-year to $36 a barrel (against a ten-year average near $22); European gasoline and diesel crack spreads

hit *all-time records*, adding roughly €0.23–0.35 a litre to European pump prices by themselves; jet fuel cracks — historically under $20 — touched $80. The five largest oil companies and five largest American refiners together earned more than $65 billion in a single quarter. Refining stocks doubled. Marathon Petroleum and Valero each rose more than 150 percent in nine months — this from SEC-filed numbers, not opinion.

Was this gouging? Legally, no — windfall is not a crime, and Part X will show exactly where the law draws the line. Economically, it was the textbook signature of a bottleneck: when the middle of a chain loses capacity faster than the ends, the middle captures the scarcity rent. The policy argument — windfall taxes, of which Portugal actually passed one at 33 percent — is a politics question taken up in Part XI. The physics question is settled: in the 2020s, the refinery, not the oil well, became the fragile heart of the price.

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PART V — MOVING IT: THE GEOGRAPHY OF PRICE

Chapter 11: Pipes, Chokepoints, and the Tyranny of Geography

Oil's journey to your tank is a geography lesson in why the same gallon costs different amounts in different places.

Inside America, the continent is carved into five administrative regions — the PADDs, a system created during the Second World War for rationing and never retired. The pattern that defines American prices is simple: the Gulf Coast (PADD 3) holds more than half of the nation's refining capacity and exports to the world; the East Coast (PADD 1) consumes the most fuel and makes the least; the West Coast (PADD 5) is an island, connected to nothing by pipeline. That is why the Gulf Coast almost always has America's cheapest gasoline and California its most expensive — before a single tax or regulation is counted.

The most consequential piece of plumbing is the **Colonial Pipeline**: 5,500 miles from Houston to New Jersey, carrying up to 3 million barrels a day of gasoline, diesel, and jet — roughly 45 percent of the East Coast's fuel, moving at three to five miles per hour, a two-to-three-week journey per batch, with storage farms along the way holding about a month of supply. Its fragility became national news in May 2021, when the Russian-linked DarkSide ransomware gang encrypted Colonial's billing systems (through an unused VPN account without multi-factor authentication) and the company shut the pipeline for six days out

of caution — the largest cyberattack on energy infrastructure in American history. Futures spiked; the Southeast panicked; at the peak, more than 16,000 stations ran dry, airlines scrounged for jet fuel, and the Biden administration declared an emergency. The company paid a $4.4 million ransom (the Justice Department recovered about half). The episode established a new category of price shock: no wells closed, no refineries broke, no ships sank — a *billing system* did.

Beyond America's borders, geography hardens into chokepoints, and the mathematics of one strait run beneath every war scare of the past fifty years. The **Strait of Hormuz** — twenty-one miles wide at its narrowest, between Iran and Oman — normally carries about a fifth of the world's oil and liquefied natural gas. The **Bab el-Mandeb** strait at the Red Sea's mouth, the **Suez Canal**, the **Turkish straits**, and the **Panama Canal** each carry smaller but critical volumes. Tankers too big for the canals round Africa's Cape of Good Hope, adding ten to twenty-one days and 40–50 percent more fuel to the journey — a hidden freight surcharge on every gallon.

There is also a peculiarly American rule worth knowing: the **Jones Act of 1920** requires that cargo moving between American ports travel on American-built, American-crewed, American-flagged ships — which cost up to three times more than foreign-flagged vessels. The result is a standing absurdity: East Coast refineries import oil from overseas while Gulf Coast refineries export American crude to Asia, because shipping Texas oil to New Jersey by sea is, by law, more expensive than shipping it to China. Economists at the University of Chicago estimated the

rule costs East Coast consumers on the order of a cent and a half per gallon and noted the revealing fact that the rule survives because its costs are spread thinly across millions of consumers while its benefits concentrate on a small shipbuilding industry. In March 2026, with the Hormuz crisis straining every supply line, the administration waived the Jones Act for oil through November — and in five months the waiver moved 41 million barrels by sea and saved an estimated $100–110 million. It was real money. It was also, as the analysts noted, a rounding error against a crude shock measured in dollars per gallon.

Chapter 12: The Invisible Hand That Stopped the Ships — Insurance as Price Maker

The most underappreciated institution in world energy is not a company or a country. It is the insurance market.

A modern tanker cannot sail without two forms of cover: hull insurance (the ship) and protection-and-indemnity cover (the liability), the latter provided by a handful of mutual clubs based in London, Norway, and Japan. If the clubs cancel, the ship is legally and commercially dead in the water — regardless of what any navy says.

This is how Iran actually closed the Strait of Hormuz in 2026 — not primarily by sinking ships, but by making them uninsurable. After the February strikes, the P&I clubs issued seventy-two-hour notices cancelling war-risk cover for Gulf transits. War-risk premiums — normally a hundredth of a percent of a ship's value — surged to between 2.5 and 5 percent, meaning $2.5 to $5 million per transit for a single supertanker. Tanker traffic through the strait collapsed from over a hundred ships a day to single digits. Hundreds of vessels, thousands of seafarers, and tens of billions of dollars of cargo simply waited.

Washington's response illustrated a theme this document will return to: the difference between announcing money and moving it. On March 11, the administration announced a $20 billion maritime reinsurance backstop through the U.S. International Development Finance Corporation and Chubb — later described in some accounts as $40 billion. By August, a Congressional

Research Service report recorded the verdict: the facility had "**sat completely idle**," because the real constraint was not the price of insurance but the fear of attack and the absence of the naval escorts it was premised on. A Lloyd's of London consortium wrote some actual business at premiums of 0.8–1.5 percent; the giant government facility wrote none. When the escort program (Operation Project Freedom) launched in May, escorts themselves never materialized at scale.

Onshore, the same war repriced refinery insurance: political-violence rates went from roughly 0.2–0.3 percent of insured value to 3–10 percent — a ten-to-thirty-fold jump — with global capacity for a single risk capped around $250–350 million, no billion-dollar policies available, and no annual Gulf war policies on offer at all: every transit was priced voyage by voyage. The September drone attack on Saudi Arabia's East-West pipeline — the one functioning bypass around Hormuz — removed four to five million barrels a day of alternative routing precisely when the strait was closed, and crude broke $100 within days. War had reached the point where the insurance market, the pipeline network, and the drone strike are three ways of saying the same thing: **chokepoint leverage equals the minimum of physical flow and insurable flow.**

Chapter 13: The Chokepoint Wars — A Field Guide to the 2026 Geography

When the strait closed, the world ran an involuntary experiment in bypass capacity, and the results are worth knowing precisely.

Saudi Arabia's East-West pipeline (Petroline) was designed for five million barrels a day and could surge toward seven, but its Red Sea terminal capped usable exports at four to four-and-a-half. In March 2026 it ran at maximum — five-point-nine million barrels through the pipe — for the first time in its history. The UAE's Fujairah pipeline (ADCOP), 1.5 million barrels a day of nameplate capacity, ran flat-out at 1.8. Iraq, whose crude flows almost entirely through the Gulf, had the Kirkuk–Ceyhan pipeline to Turkey's Mediterranean terminal with 1.5 million barrels a day of theoretical capacity — but after years of arbitration disputes and decay it was delivering only 170–250 thousand, supplemented by an improvised fleet of four hundred tanker trucks a day shuttling Basra crude north. Iran's own "bypass" to its Gulf-of-Oman terminal at Jask existed mostly on paper.

Add it up: roughly three-and-a-half to five-and-a-half million barrels a day of bypass against a strait that normally moved about twenty-one. When Iraqi drones struck the East-West pipeline's pump stations in September, even that partial answer went offline for weeks. The arithmetic was the story of 2026 in one line: **the floor of the crisis was set not by diplomacy but by pipe diameters.**

And the coda: within weeks of the crisis, Iraq approved a five-billion-dollar pipeline to finally connect its southern fields to

its northern export routes, the UAE began building a second pipeline to double Fujairah's capacity, and Saudi Arabia accelerated terminal upgrades. Chokepoint dependency had been repriced from a market inconvenience to a sovereign-risk variable — the first time in the industry's history that geography itself appeared as a line item in national budgets.

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PART VI — THE STACK AT THE PUMP

Chapter 14: Anatomy of a Gallon

Now the question this entire document exists to answer: when you pay $4.37 at the pump, *who gets what?*

The Energy Information Administration divides the retail price of gasoline into four components, and in the autumn of 2026 they stacked like this:

- **Crude oil: about 57 percent.** The oil itself — the inheritance from Chapter 1, the benchmark prices from Chapter 5, the wars from Part XV. When you blame "the price of oil," this is the share you're blaming. - **Refining: about 21 percent.** The kitchen from Chapter 9 — normally 14–20 percent of the price, pushed to its historical ceiling by the 2026 crisis and the capacity story from Chapter 10. - **Taxes: about 14 percent in the U.S. average** — 18.4 cents federal (unchanged since October 1993), plus a state average of about 33.6 cents, for roughly 52 cents a gallon. But "taxes" is not one thing — the chapters below will unpack the federal layer, the state layer, the hidden swipe-fee layer, the biofuel-credit layer, and the carbon layer that stack on top of each other like sediment. - **Distribution and marketing: about 8 percent** — the pipelines, trucks, stations, and the thin retail margin on which the next part of this document dwells.

Those shares move. In 2022 the crude share hit 61 percent; in 2026 the refining share ballooned. The anatomy is a living thing, and Part XV tracks it through the strangest year in the modern history of the number.

Chapter 15: The Tax Stack — From 18.4 Cents to Policy Architecture

Begin with the number that hasn't changed since 1993: 18.4 cents a gallon federal tax (plus a tenth of a cent for a leaky-underground-tank fund), funding the Highway Trust Fund. Because it is a fixed number rather than a percentage, inflation has quietly halved its real value over three decades — the highway fund has been running on general-fund transfusions since 2008, over $275 billion of them, and the Congressional Budget Office projects insolvency by 2028 with a decade deficit around $340 billion. The tax built the interstate highway system; the electric vehicle (EV) is unbuilding its revenue logic, and states are improvising fixes — flat EV registration fees, mileage-based pilot programs, and in Arizona a proposed constitutional amendment to ban mileage tracking altogether. The 18.4 cents is in the price today; it may not be in the anatomy much longer.

Now the states: from about 15 cents in Alaska to California's stack — a 61.2-cent excise, the federal tax, a 2-cent tank fee, roughly 2.25 percent sales tax, and two programs no other state has: a cap-and-trade carbon cost and a low-carbon fuel standard, together adding an estimated 40–50 cents a gallon by 2026. California's full policy stack runs 80–90 cents a gallon — the difference between California's $6-plus pump price and Texas's under-$4 in a single number. Whether that premium is worth its environmental return is one of America's oldest policy arguments; this document records the stack, not the verdict.

Now leave America. In the United Kingdom, fuel duty of 52.95 pence per litre plus 20 percent VAT — a tax *on* the tax — means roughly half of every pound at the pump is government revenue; Germany's energy tax, carbon price, and VAT run about 42 percent of the pump price. In Norway, the fuel excise approaches $1.82 a litre. And at the other end of the world map sit the subsidy states: Libyan motorists pay about nine cents a gallon, Iranians about eleven, Venezuelans about thirteen — prices that have nothing to do with the cost of oil and everything to do with political choices, and that funnel a tenth to a sixth of some national budgets into keeping fuel cheap. The world map of gasoline prices is a map of governance: the hundred-fold spread between the cheapest and most expensive gallon on Earth is set, in the words of one careful study, "almost entirely by government policy, not by where the oil comes from." An American pays close to the untaxed cost of fuel; a European pays double as a deliberate fiscal and environmental choice; a Venezuelan pays a hundredth as a deliberate political one.

Chapter 16: The Hidden Layers — Swipe Fees, Biofuel Credits, and the Boutique-Blend Economy

Between the big components of the pump price sit three layers most drivers never see.

The swipe-fee layer. Roughly a third to half of American fuel purchases go on credit cards, and card networks charge retailers a percentage of the *entire* transaction — including the taxes. As pump prices rise, the fee per gallon rises automatically: by mid-2026, with regular above $4.40, the card cost worked out to roughly 10 cents per gallon — and about 13 cents on $6 diesel. This is a hidden multiplier with a perverse property: it scales with the crisis. The retailers' trade groups count record swipe fees nationally ($198 billion in 2025) and note gasoline retailers now pay more in card fees than they earn in profit. The contrast with Europe is striking: EU law caps card interchange at 0.3 percent — the American rate is seven times that. No single actor sets out to raise your gas price by a dime; the percentage just does what percentages do.

The biofuel-credit layer. America's Renewable Fuel Standard, created in 2005 and expanded in 2007, requires refiners and importers to blend rising volumes of ethanol and biodiesel into the fuel supply, or buy compliance credits — Renewable Identification Numbers, or RINs — from those who do. In 2026, the obligation equaled roughly 15.5 percent of every gallon sold. RINs are a traded commodity, and in 2026 they doubled to about $2.40 — translating, by various estimates, into

somewhere between the EPA's official estimate of five cents per gallon of gasoline and an industry-aligned model's estimate of 37–45 cents. (Both frames are on record here; the EPA's treats the credit as a cross-subsidy that nets out across the supply chain, the industry models treat it as a pure pass-through cost.) The politics are a permanent war: small refineries win exemptions, biofuel states demand the volumes be reallocated, courts vacate denials, and the whole apparatus whipsaws with each rulemaking — a visible, quantifiable demonstration of how a Washington docket can move a pump price by tens of cents within a year.

The boutique-blend layer. Gasoline is not one recipe. Summer gasoline must meet a federal vapor-pressure cap (RVP) to prevent smog — which pushes out cheap, high-vapor butane and adds an estimated 10–30 cents a gallon for the summer recipe. More than a dozen regional blends exist, and California's unique CARB formula is made by so few refineries that a single plant outage can move the whole state's price — which is exactly what happened in 2026. (The E15 saga belongs here too: ethanol's chemistry shows 15 percent blends perform identically to 10 percent on vapor pressure. But a one-psi line written into the 1990 Clean Air Act still bans E15 in summer across half the country absent waivers — the kind of statutory fossil that turns into a recurring political ritual.) Add the legacy layers — the reformulated-gasoline rules of 1990 (about seven cents a gallon at introduction), ultra-low-sulfur diesel (three cents), double-hull tankers after the Exxon Valdez (about sixteen cents per barrel of freight) — and the pattern emerges: **each generation's environmental bargain with the internal combustion engine**

is permanently embedded in the pump price, a few cents at a time, compounding.

Stack it all up for a California driver in the autumn of 2026: the oil, the refining, the federal and state taxes, the carbon programs, the boutique blend, the card fee, the credit cost — and you arrive at a price that is less a market outcome than a sedimentary record of thirty years of policy geology.

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PART VII — SELLING IT: THE LAST THIRTY FEET

Chapter 17: The Station — America's Most Misunderstood Business

Pull into any American gas station and you are looking at one of the thinnest-margin businesses in the country, wrapped around one of the most profitable.

The fuel itself earns the station very little: a gross margin of perhaps 30–40 cents a gallon, from which must come card fees (Chapter 16), freight, labor, rent, insurance, compliance, and the occasional environmental liability — older stations carry underground-tank cleanup risks averaging well into six figures per incident, a hidden legacy cost that has helped push the national station count down from over 200,000 in the 1990s toward roughly 148,000 today. Net, a station earns pennies per gallon. The real business is inside: the convenience store, where margins are fat and the fuel was merely the bait that brought you. Roughly 80 percent of American fuel is now sold by convenience stores.

The industry's history is a tour of American commerce. The first drive-in station opened in St. Louis on December 1, 1905 — a gravity-fed tank and a garden hose. Early pumps were glass, so drivers could see the measure and trust it. By the 1920s there were 300,000 outlets; by the 1970s, "gas wars" — below-cost price cutting between neighbors — were a recognized phenomenon with their own legal doctrine (major companies'

targeted price support to favored dealers during wars triggered Robinson-Patman discrimination cases in the 1950s). New Jersey banned self-service in 1949, Oregon in 1951; Oregon lifted its ban in 2023, leaving New Jersey the last full-service holdout, where polls show three in four residents want it kept.

Two modern innovations reshaped the last thirty feet. The first is the warehouse-club model: Costco, which entered fuel retail in 1995, prices gasoline near cost — often 20 to 70 cents below competitors — as pure traffic bait for the warehouse, and has grown to more than 740 stations, roughly a tenth of national fuel volume, gaining market share in every price spike. The second is the digital layer: price apps like GasBuddy (crowdsourced prices from 150,000 stations) and cashback products, where the discount of 5–25 cents a gallon is typically split between the brand buying traffic and the platform arbitraging payment rails and data. The apps redistributed pennies; they did not change the stack underneath.

And a caution from the research: fuel-price transparency is one of the most contested ideas in applied economics. When *consumers* get real-time price data, competition intensifies — studies of highway price boards and app adoption show lower prices. But when *retailers* get real-time views of each other — and algorithms can respond in milliseconds — the opposite can happen: several studies of European transparency regimes found price *increases*, as algorithms learn that undercutting will be instantly matched, and one Australian market took twelve years of transparency to settle into a stable coordinated structure among six competitors. Germany, in 2026, became the first major

economy to rate-limit the process, allowing stations to change prices only once a day. The modern "gas war" is fought by algorithmic pricing, at machine speed — and regulators are still writing the rules of engagement.

Chapter 18: Rockets and Feathers — Why It Rises Fast and Falls Slow

Every driver knows the phenomenon: the station's sign changes overnight when crude jumps, but takes weeks to follow crude back down. Economists call it "rockets and feathers," and it is one of the best-studied patterns in all of price economics.

The evidence, fairly stated: the landmark 1997 study of Borenstein, Cameron and Gilbert found retail gasoline responds to crude-oil *increases* in about four weeks but to decreases in about eight, and — critically — located the asymmetry at the *retail* stage: refiners and wholesalers showed no such pattern. The Federal Trade Commission's own economists confirmed it (prices rising more than four times as fast as they fall, concentrated in branded stations and concentrated markets, varying year to year). Skeptics exist — a respected line of research finds no asymmetry in the crude-to-wholesale link, and one influential model shows search-cost-driven consumer inattention can produce the pattern in a market with *zero* collusion.

The honest synthesis for the reader: the asymmetry is real and robust at the rack-to-pump step in most studies; its cause is genuinely contested — market power, consumer inattention, and inventory economics all have evidence behind them; and no study adopts the political conclusion that any particular company is to blame. Britain's competition regulator reached the same empirical conclusion from the other side in 2023: no cartel — but supermarket fuel margins had risen six pence a litre in three years, the historical price-leader — the supermarket chain Asda —

was caught "feathering" diesel prices on the way down, and the remedy chosen was a transparency law compelling every station to publish prices in real time. Asymmetry, like beauty, is in the eye of the dataset.

What is certain is the mechanism's *effect*: in every crisis, the pass-through into your tank is quicker than the relief on the way back down — an asymmetry written into the rhythm of every price cycle this document describes.

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PART VIII — THE MONEY TRAILS

The chapters that follow map the documented flows of money through the oil system: the bribes, the thefts, the manipulations, the evasions — and the record of those who prosecuted them, including the times the prosecution itself broke the law. Every name in this part appears in court records, government enforcement releases, or formal indictments. Where a case ended in acquittal or dismissal, that ending is stated with the same prominence as the accusation. Where a claim remains unproven, it is labeled as such. This part is not an indictment of an industry; it is an anatomy of an incentive.

Chapter 19: The Trader Docket — Bribery as a Business Model

Between the national oil company that sells a cargo and the refinery that buys it stands a species of firm most drivers have never heard of: the independent commodity trader. Vitol, Glencore, Trafigura, Gunvor, Freepoint — private partnerships in Geneva, London, Amsterdam, Stamford — and the largest movers of the world's physical oil. From 2016 to 2024, the U.S. Department of Justice and the CFTC ran the most consequential anti-corruption sweep in the history of this layer, and the results, now final, deserve to be read as a set, because the pattern repeats with remarkable fidelity.

Vitol. In December 2020, Vitol became the subject of the first-ever coordinated Justice Department–CFTC foreign-bribery resolution. The company admitted paying more than $8 million in bribes to at least four officials of Brazil's Petrobras in exchange for — this phrase matters — *confidential information including the prices competing traders had bid*: insider trading on government auctions, executed with cash, shell companies in Curaçao and Panama, fake consulting agreements, and code names including "Batman," "Popeye," and "Phil Collins." A second scheme funneled over $2 million to Mexican and Ecuadorian officials. Penalties totaled about $164 million. Vitol's former Houston trader Javier Aguilar was convicted by a jury in 2024 and sentenced to four years in prison; seven co-conspirators pleaded guilty, three of them foreign officials.

Gunvor. In March 2024, the Geneva-based trader pleaded guilty to a bribery conspiracy in Ecuador: nearly $100 million routed to intermediaries between 2012 and 2020, knowing portions would reach Petroecuador officials who steered oil-backed loan contracts to front companies — one grateful official received an 18-karat gold Patek Philippe. The price was a $374 million criminal fine, $287 million forfeiture, plus parallel Swiss and Ecuadorian payments exceeding $90 million more. Gunvor had already been convicted in Switzerland in 2019 — a first for a major trader — for bribes in Congo-Brazzaville and Ivory Coast.

Trafigura. In March 2024, the largest of the independents pleaded guilty to paying roughly $19.7 million in bribes to Petrobras officials, earning more than $61 million in profits. Notably, the Justice Department gave Trafigura only minimal cooperation credit, criticizing its slow document production and slow discipline of implicated employees. A separate CFTC action in 2024 added $55 million for trading gasoline on non-public information misappropriated from a Mexican state-linked entity and for manipulating a fuel-oil benchmark. (The company's own 2025 annual report disclosed an internal fraud of $1.1 billion in its Mongolian oil business, discovered in 2024. It is a reminder that the traders are cheated too.)

Glencore. The largest case of all: in May 2022, Glencore pleaded guilty to a bribery conspiracy spanning 2007–2018 across Nigeria, Cameroon, Ivory Coast, Equatorial Guinea, Brazil, Venezuela, and Congo — over $100 million paid through intermediaries — with the Justice Department memorably stating

that "bribery was built into the corporate culture." The same month, the CFTC fined Glencore $1.186 billion for manipulation and corruption, the largest penalty in the agency's history. Britain's Serious Fraud Office secured guilty pleas on seven counts, calling the conduct "endemic." A court-appointed monitor was installed — and terminated fifteen months early in March 2025, without public rationale.

Freepoint. In December 2023, Stamford-based Freepoint paid more than $91 million to settle charges that from 2012 to 2018 it bribed employees of a South American state oil company — again, Petrobras — for fuel-oil production plans and, once more, *competitors' bid prices*. Its trader Glenn Oztemel was convicted in October 2024.

Step back and the structure of the whole layer becomes visible. The same techniques recur for two decades across five companies and three continents: shells, fake invoices, alias emails, code names, gifts. The bribe purchases a specific, monetizable asset — *discretion*: the award of a tender, the timing of a cargo, the competitor's number. The victim of the information theft is every honest bidder, which means ultimately the selling state and, in the case of subsidized fuel systems like Brazil's or Ecuador's, the public treasury. This is why the document places this chapter directly before the politics chapters: the corruption is not an aberration on top of the price system — at this layer, it *is* part of the price formation, a tax paid to access government discretion, invisible in any pump-price decomposition.

Chapter 20: The Other Half of Every Bribe — The NOCs

A bribe requires a bribee. The national oil companies — PDVSA of Venezuela, Petrobras of Brazil, Pemex of Mexico, and their kin — appear in the docket not as background but as co-conspirators whose officials were the network's load-bearing members.

PDVSA holds the record for sheer scale of the American prosecutions: roughly two dozen individuals charged, most pleading guilty, in schemes that included supply-contract kickbacks, "pay-to-play" fees to collect past-due invoices from the very state oil company, and embezzlement through currency-allocation programs worth hundreds of millions. One treasurer laundered $160 million through Florida accounts; a procurement chief at Citgo accepted World Series tickets and Miami condos. The social collapse of Venezuela's oil sector — production falling by three-quarters from its peak — has many causes, but the docket establishes that insiders were looting the machine in parallel with its political destruction.

Petrobras and Odebrecht form the largest bribery scheme ever documented in the oil chain: overpriced contracts, a literal corporate "Department of Structured Operations" inside the construction giant Odebrecht running an off-book bribery budget through Caribbean shells, more than $3.5 billion in combined U.S. penalties, and political aftershocks that felled a presidency (Dilma Rousseff's impeachment) and jailed a former president (Luiz Inácio Lula da Silva — whose conviction was later annulled, a thread we return to in Chapter 24).

Pemex contributed the most politically radioactive single case: its chief executive Emilio Lozoya — arrested, charged with accepting $10.5 million from Odebrecht, a slice of which he says funded President Enrique Peña Nieto's 2012 campaign, testifying that he acted on the president's orders. The evidentiary rulings hobbled the case; it remains unresolved, and the allegations against Peña Nieto remain unproven. Mexico's 2017 "gasolinazo" — the 20 percent price-liberalization shock that ignited nationwide looting and protests, detailed in Chapter 33 — belongs in the same sentence: a fuel-price system rotted by procurement corruption met its public in a single announcement.

PetroSaudi, finally, shows the purest form of the archetype: a fake joint venture with Malaysia's 1MDB sovereign fund in which $1 billion of public money bought a 40 percent stake in a company whose claimed crown-jewel Turkmenistan field interest it had never actually owned — a terminated contract it repackaged. The scheme's architects, Tarek Obaid and Patrick Mahony, were convicted by a Swiss court in 2024 and ordered to repay roughly $2 billion; the fugitive financier Jho Low remains at large, and over a billion dollars has been recovered by the U.S. Justice Department from the wider looting.

Chapter 21: The Block-Award Scandals and the Men in the Middle

Two further network types deserve their own map, because they sit at the wellhead where value is greatest.

The OPL 245 affair (Nigeria). In 1998, Nigeria's military ruler awarded a nine-billion-barrel offshore block to a company incorporated five days earlier, secretly controlled by his own petroleum minister, Dan Etete — convicted of money laundering in France in a separate case. A decade later, Shell and Eni paid $1.3 billion for the block, routed through a London escrow at JPMorgan; within days, over a billion flowed to the front companies. What happened next is the instructive part: in Milan, after a marathon trial, Shell, Eni, and their executives — including Eni's CEO — were *acquitted* in 2021 for insufficient evidence; Nigeria's own case against its former attorney-general collapsed; but in 2024, two of the *Italian prosecutors* were themselves convicted for concealing exculpatory evidence from the defense. The block, twenty-eight years on, remains undeveloped; in 2026 the Nigerian government split it four ways and handed the pieces back to the incumbents, and the exiled "Malabu" faction now sues the state for a trillion naira. Every institution that touched this file — company, official, bank, prosecutor on both sides — left a record. The document presents the record without a verdict the courts themselves never reached.

Unaoil maps the intermediary layer — the "consultants" who sit between Western contractors and oil ministries. The Monaco-based firm, run by the Ahsani family, was the middleman

of choice for contracts in Iraq, Kazakhstan, and Angola, passing over $17 million in bribes and enabling $3.6 billion in contracts for clients including Technip and Samsung. The family pleaded guilty in Texas; three associates were convicted in London. And here the investigators themselves enter the record: three of those London convictions were *quashed on appeal* after it emerged that the head of the Serious Fraud Office had held improper contacts with a private investigator working for the Ahsanis — an official review condemned the leadership's "gullibility and arrogance." The guilty pleas in America survived; the British cases died. Same facts, two justice systems, opposite outcomes.

The services giants complete the wellhead map. The TSKJ consortium — KBR (then Halliburton), Technip, Snamprogetti (Eni), and Japan's JGC, with Marubeni as agent — bribed Nigerian officials across three presidential administrations to win $6 billion of LNG plant contracts, using a Gibraltar front company, a London lawyer named Jeffrey Tesler, briefcases of cash, and bullion vans. The American resolutions totaled over $1.7 billion (KBR's $579 million was then the largest FCPA settlement ever); Tesler forfeited $149 million. Nigeria's anti-graft agency, for its part, filed charges naming KBR's then-CEO — and, in a headline that circled the world, *former U.S. Vice President Dick Cheney*, who had led Halliburton in the relevant years. The charge was never prosecuted; the EFCC raided Halliburton's Lagos office and extracted a settlement. Baker Hughes paid $44 million over Kazakhstan bribes routed through an Isle of Man "consulting" firm; Chevron paid $30 million for disguised kickbacks under the UN's Iraq Oil-for-Food program.

The reader should notice what these networks share: **the bribe is always the price of a gate that a human controls** — a license, a tender, an allocation, an inspection. Wherever oil money passes through such a gate, the docket shows, someone priced the key.

Chapter 22: The Manipulation Docket — Where the Law Draws the Line

Part VIII's opening promised to show exactly where lawful windfall ends and unlawful manipulation begins. The answer is not a theory; it is a stack of court files.

The proven cases are all *microscopic*: squeezes and games at specific market moments. In 2008, the firms Parnon and Arcadia amassed a dominant position in physical crude at Cushing — the delivery point from Chapter 7 — then squeezed the spread and dumped into the daily pricing window; the CFTC extracted $13 million and trading restrictions. In 2007, Optiver attempted to manipulate the *closing prices* of crude, heating oil, and gasoline futures — settled for $13 million plus a million in disgorgement; the complaint's pages of contemporaneous trader chats are permanently on record. In 2020, Vitol attempted to manipulate two Platts fuel-oil benchmarks (part of the same resolution as its bribery case); in 2018, a Swiss firm settled charges of leaning on the European gasoline benchmark with discounted physical sales while short the futures — though a sitting CFTC commissioner publicly dissented from the enforcement theory. Glencore's billion-dollar penalty covered "spoofing" — placing orders with no intention of executing them, to fake supply and demand — across oil and metals over more than a decade.

The defense bar's wins matter just as much for defining the line: the CFTC *lost* its first litigated insider-trading case against EOX Holdings (the Fifth Circuit citing lack of fair notice) and lost the 2018 DRW case, where a judge found the accused trader's

aggressive-but-honest bidding was bona fide and the price not artificial. The boundary the courts have drawn: deception, artificial prices, and breached duties are illegal; conviction-scale position-taking, even at market-moving size, is not.

This distinction — microscopic manipulation proven, systemic manipulation unproven — runs through the whole document. When a future chapter shows speculators *net short* at the 2026 price peak, that is not a vindication of traders; it is the market working as designed. The law polices the cheats at the margin of the game, not the game's existence.

Chapter 23: The Parallel Petroleum Economy — Theft, Fraud, and the Anti-Theft Rent

Nowhere is the two-sided nature of oil money more visible than in Nigeria, where a measurable theft network and a controversial anti-theft contract coexist, and where the official record itself has opened a new file.

The theft network, by the numbers. At its peak around 2021, an estimated 102,000 barrels a day — over a tenth of national production — was being stolen through illegal pipeline taps and creek refineries, feeding both adulterated local fuel and export cargoes moved by ship-to-ship transfer; legislators at the time cited losses up to $10–15 billion a year. The economics are simple: subsidized domestic fuel created an arbitrage — buy cheap in Lagos, sell at world price across the border — and theft industrialized the arbitrage. Then two things happened in sequence: the fuel subsidy was abolished in 2023 (removing the arbitrage's root), and the state awarded a private pipeline-surveillance contract — for roughly four billion naira a month — to Tantita Security Services, a company led by Government Ekpemupolo, known as Tompolo, a former militant commander whose 2016 arrest warrant on a separate 40-count fraud charge (filed by the anti-corruption agency, never resolved). That history coexists with a data-supported result: measured theft fell more than ninety percent, to its lowest since 2009, and national production nearly doubled from its 2022 trough. Nigeria's legislature formally endorsed the arrangement in 2026.

Critics — including civil-society petitioners and rival ex-militant figures — allege the security corridor was carved among connected firms at a combined cost approaching $1.3 billion a year and question who surveils the surveillors; the defenders point at the production curve. Both are on record. The document notes only the structural pattern: **the same networks often stand on both sides of the theft/anti-theft line, and the state repeatedly converts crisis into contract.**

The paper fraud. The subsidy era left a companion record: "ghost" import claims for fuel that never arrived, over-invoicing, forged bills of lading, and single cargoes claimed multiple times — what a former anti-corruption chief called systemic collusion between officials and importers, with some estimates of subsidy-era leakage running to many billions of dollars a year. And in April 2026, a committee of Nigeria's own House of Representatives opened a formal inquiry into allegations that international and indigenous companies had been *buying Nigerian crude below market price* — crude underpricing as a transfer from the treasury to the buyer. The investigation was ongoing at this writing.

The loan stack. The same barrels are financially committed multiple times forward. In 2023 Nigeria's state company pledged 164 million barrels of future crude — 90,000 barrels a day from 2024 — to repay a $3.3 billion emergency loan arranged through Project Gazelle, a Bahamas financing vehicle arranged by the African Export-Import Bank and syndicated with Vitol and Sahara Energy; the stack was refinanced in 2026 through a $5 billion swap with First Abu Dhabi Bank and a further $4.5 billion federal

refinancing. Every pledged barrel is a barrel unavailable for the domestic refinery program (Chapter 40) or for export. The oil-backed loan — the state selling its inheritance forward at a discount — is the sovereign twin of the trader-bribe circuit, and its beneficiaries are named in the loan documents.

Chapter 24: Investigating the Investigators

A non-biased anatomy must also map the failures of the mapping institutions, and this record is substantial.

The collapse of Lava Jato. Brazil's Car Wash investigation — the largest anti-corruption operation in history, which convicted a former president and dozens of oligarchs — was itself annulled in large part by Brazil's Supreme Court, which found that the star judge, Sergio Moro, had colluded with prosecutors (as revealed by leaked messages), voided the pivotal plea agreements, and suspended billions in fines. The bribery it uncovered was real; so was the due-process violation that destroyed the convictions. The document holds both facts.

The Unaoil quashals and the Milan convictions. As Chapter 21 recorded, three British convictions died over improper contacts between the SFO's leadership and a private investigator; a government review condemned the office's conduct. Meanwhile in Milan, two prosecutors were convicted for hiding exculpatory evidence from Eni's defense in the OPL 245 trial — misconduct that retroactively shadows even an acquittal.

The FCPA pause. In February 2025, a U.S. executive order suspended Foreign Corrupt Practices Act enforcement, with new guidelines narrowing it to cases touching national interest and shifting emphasis to individual accountability; the first corporate declination under the new policy came in March 2026. Transparency advocates called it the return of bribery as routine business; enforcement officials called it refocusing. Meanwhile — and this is the pattern the next chapter develops — sanctions

enforcement against Iran and Russia *intensified* under the same administration. The two arms of financial law enforcement moved in opposite directions, and both movements are documented.

The whistleblower layer. Behind several of this part's biggest cases stands a person who risked everything to expose them: the Vitol case was built on a whistleblower's account; the PetroSaudi file opened when Xavier Justo, the company's own co-founder, stole the data that proved the fraud and handed it to journalists — for which he was himself prosecuted in Thailand; and Lava Jato's exposure of its own judge's collusion came from reporters working with leaked messages. Every enforcement era in this document was, in the end, started by an insider. The CFTC and SEC run award programs that pay whistleblowers a share of recovered penalties, and those programs, more than any statute, now set the docket's agenda.

The doctrine wars. CFTC Commissioner Caroline Pham publicly dissented from her own agency's benchmark-manipulation case; the agency lost its first litigated insider-trading prosecution on fair-notice grounds; and Congress, in 2026, began legislating on markets that did not exist when the statutes were written. The machinery of enforcement is itself contested terrain — which is precisely why this document treats court outcomes, not press releases, as the load-bearing evidence.

Chapter 25: The Gray Circuit — Sanctions, Shadow Fleets, and One Platform, Three Cargoes

Since 2019, a gray market has grown up around the sanctioned barrels of Iran, Russia, and Venezuela — and the documentation now allows it to be mapped as a single platform serving three cargoes.

The architecture, reconstructed from U.S. Treasury designations and financial-intelligence reports: sanctioned crude loads at Kharg Island or a Russian Baltic port; a "shadow fleet" of more than a thousand aging tankers — roughly a fifth of the world's capacity — with opaque ownership, flags of convenience, and transponders that go dark; cargoes blend at sea (the "Malaysian blend" relabeling Iranian crude as Malaysian became so routine that it distorted Malaysia's own trade statistics); payments route through Hong Kong shell companies, UAE free-zone fronts, and cryptocurrency — one Treasury-documented network moved over $600 million in a year, and an exchange-house ecosystem in Tehran and Dubai launders the proceeds. The buyers of last resort are a ring of roughly forty independent "teapot" refineries in China's Shandong province, paying in yuan through small banks.

The named nodes form a network map in the archive: the Shamkhani family's Wellbred companies (which bought a French cooking-oil refinery as a legitimacy layer); the Tsoris shipping-services cluster, the 2Rivers network designated by Britain in February 2026; the Indian, Emirati, and Chinese ship

managers; the Turkish gold-and-crypto broker; the British Virgin Islands and Hong Kong shells. The enforcement record is real — five hundred Russia-related designations by early 2026, Britain's first criminal prosecution of a shadow-fleet captain, record fines for sanctions circumvention — and so is the adaptation: the platform hardened in exactly the ways enforcement taught it to.

Two structural observations close the chapter. First, the discount is the price of the law: the gap between Brent and sanctioned Urals or Iranian crude — which swung from a record $40 discount to *premiums* during the 2026 crisis — is the market clearing the cost of legal risk, and it is large enough to fund the entire evasion apparatus with room to spare. Second, the platform proved to be wartime infrastructure: when the 2026 war began, it was the gray circuit — teapots, dark tankers, yuan settlement — that kept Iranian crude moving at half rate while the formal economy around it shut down. The law built the shadow market; the war stress-tested it.

Chapter 26: The 2026 War-Trades Docket — The First War With an Audit Trail

No previous war in history left a real-time, public, on-chain record of who bet what before the bombs fell. The 2026 war did, and the docket that resulted deserves its own chapter.

The prediction-market evidence. Academic analysis of the full Polymarket dataset — the Mitts & Ofir study of more than a million accounts — identified more than 210,000 suspicious wallet-market pairs with an aggregate win rate near 70 percent and roughly $143 million in anomalous profit. In the Iran-strike markets specifically: six wallets funded within twenty-four hours of the February 28 strikes earned about $1.2 million; one trader — first mapped publicly by the blockchain-analytics firm Bubblemaps — ran a documented 93 percent win rate across three successive military events (October 2024, June 2025, February 2026), preloading wallets days in advance; a cluster of thirty-eight linked accounts cleared over $2 million on the strike date alone. The platforms cooperated: Polymarket flagged and referred; Kalshi had blocked the most famous defendant's know-your-customer checks days before his trades.

The first prosecution. In April 2026, federal prosecutors in Manhattan indicted Army Special Forces Master Sergeant Gannon Van Dyke, a planner of the January operation that captured Venezuela's Nicolás Maduro, for betting $33,000 on prediction markets that Maduro would fall — profit, $409,881 — using classified operational knowledge. The charges deploy statutes never before used: the "Eddie Murphy Rule," a

Dodd-Frank provision named after the 1983 film *Trading Places*, in which the Duke brothers bribe a government clerk for the orange-crop report — the law that finally made it a crime for federal personnel to trade commodities on confidential government information. The government's classified-evidence procedures are active; trial is set for December 2026. A second, template-setting case followed: Google engineer Michele Spagnuolo, charged in May 2026 for trading on stolen internal search-trend data — establishing that the doctrine reaches corporate information, extraterritorially. A third extended it again weeks later: criminal charges over an employee misappropriating an employer platform's confidential information to trade on a crypto exchange — each case widening the map of whose information the law will now protect.

The bigger, open case. The futures-market evidence is far larger and far less resolved. Reuters-documented analyses count roughly **$7 billion of suspicious short positions** in crude, gasoline, and diesel futures across four windows timed minutes to hours before major war announcements — including a single pre-announcement minute in March that moved twenty thousand lots. The Justice Department is investigating; the exchanges are investigating; no futures trader has been charged, and the legal experts quoted on record call the volumes "highly unusual... concentrated." A chorus of allegations from political figures — including claims of hundreds of millions in profits from trades ahead of the strike moratorium — remains unverified, and this document records them as claims, with the official denials alongside. Congress has responded with a stack of bills to ban

trading on classified information in prediction markets.

The pattern worth flagging — the same pattern Chapter 6 established for 2008 — is emerging again: the *positioning* record contradicts the simple story of speculators driving the spike, while the *timing* record around specific announcements justifies the open investigation. Both truths fit in the same market. The audit trail exists; attribution is the unfinished work.

Chapter 27: Venezuela, 2026 — The Newest Oil State, Under New Management

On January 3, 2026, U.S. special operations forces captured Nicolás Maduro in Caracas; an interim government under Delcy Rodríguez took over the country holding the world's largest proven reserves. The oil settlement that followed is the newest entry in this part's ledger, and every element of it is documented.

Within days, the Department of Energy announced the sale of 30–50 million barrels of Venezuelan crude, with proceeds routed to a **U.S.-supervised fund in Qatar**; the interim government announced its first $300 million transfer within weeks. Two trading houses — Vitol and Trafigura — were granted licenses and effectively tasked with marketing the program. A Vitol trader captured the arrangement's logic on the record at a January White House meeting with the majors: "*We're here to ensure that you're going to be able to move all of this oil all around the world at the best price possible so that the influence that you have over the Venezuelans will ensure that you get what you want.*" Exports recovered from under 200,000 barrels a day during the blockade to over a million by spring; the United States became the destination of 45 percent of Venezuelan crude — the highest since 2019 — with India second.

The follow-the-money threads, flagged rather than resolved: the governance and audit terms of the Qatar fund are unpublished; the traders' fees on the mandate are unpublished; the pre-capture gray-market discount of about $15 a barrel compressed to about $5 — value that previously accrued to

shadow intermediaries now accruing somewhere inside the licensed channel, to parties not publicly identified; and the lobbying firm most closely associated with the administration — Ballard Partners, registered at $60,000 a month for Saudi Arabia — opened a Caracas office in July 2026. All of this is legal; all of it is documented; none of its beneficiaries are fully visible. The reader is left with the same exercise as in every chapter of this part: a map, not a verdict.

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PART IX — THE PATTERNS

Fifteen patterns recur across this document's sixteen research dives. Each is stated with its mechanism, its evidence, its counter-evidence, and its open questions. Patterns are not accusations; they are regularities awaiting explanation.

Chapter 28: Fifteen Patterns

Pattern 1 — The Waiver Oscillation. Sanctions tighten; a crisis erupts; a temporary general license opens a window; sanctioned barrels flood through; the window closes; repeat. In 2026 alone the Treasury issued, revoked, and reissued five licenses across Iran and Russia in four months — including GL X in June, the most expansive Iran license since 1979, revoked after fifteen days. Quantified through the windows: tens of millions of barrels moved, with Iran's net gain estimated by analysts at roughly $11 per barrel (discount compression plus avoided shadow costs). The academic verdict on the era — Stanford's study titled *The Myth of Maximum Pressure* — is that Iran's exports exceeded their nuclear-deal-era peak by 2024, through a combination of deliberate relaxation and maturing evasion. All angles on record: the valves work as release mechanisms; the reversals breed cynicism (one Treasury reversal came within seventy-two hours of a public refusal to renew); each window upgrades the evasion platform; and the beneficiaries are measurable (Iran, Russia, Indian refiners, pre-positioned inventory).

Pattern 2 — Announced Money vs Moved Money. The 2026 crisis produced a consistent gap between the scale of announced relief and its mechanical effect: the $20–40 billion reinsurance backstop that wrote zero coverage; the escort program whose escorts never materialized; strategic reserve releases covering weeks of a shock measured in months; excise holidays an order of magnitude smaller than the price move (real but tiny: the Jones Act waiver's $100 million saved against a dollar-a-gallon crude

shock). One documented angle: governments optimize for visibility under uncertainty. Another: the binding constraints are physical and cannot be waived. A third, worth a writing project of its own: each failed lever teaches markets to discount the next announcement.

Pattern 3 — The Triple-Committed Barrel. Nigeria's same physical crude is pledged simultaneously to bank repayment (90,000 barrels a day), the domestic-supply program (450,000 a day, of which barely half arrived), and export contracts. Delivery failures under one commitment are the arithmetic consequence of the others. The pattern generalizes: wherever a resource is the collateral for the system that consumes it, scarcity is manufactured on paper before geology ever weighs in.

Pattern 4 — One Platform, Three Cargoes. The sanctions-evasion architecture (UAE and Hong Kong shells, dark tankers, ship-to-ship blending, crypto rails, Shandong teapots) serves Iran, Russia, and Venezuela interchangeably, with shared facilitators. Chapter 25 mapped it; the pattern is that sanctions built a single gray market rather than three separate ones.

Pattern 5 — The Bribe as Price Input. From Vitol's purchased bid information to Freepoint's stolen competitor numbers: at the trader–state interface, confidential information is bought and sold as a commodity, and the purchase price is invisible in every pump-price decomposition. The corollary: countries where this layer is thickest display a governance premium at the pump — Nigerian fuel was historically among the world's most expensive relative to income in an oil-exporting nation.

Pattern 6 — The Enforcement Pendulum. Crackdown (2016–2024: six corporate resolutions, over twenty individual convictions, $1.7 billion-plus) → pause (the 2025 FCPA suspension) → narrow resumption (2026: individuals-first, national-interest filter, first corporate declination). Meanwhile the monitorship of the largest offender was terminated early. No new oil-trader FCPA case has appeared since the regime change; correlation, not established causation.

Pattern 7 — The Insider-Information Regime Shift. For the first time, a war produced a public audit trail of pre-event betting; for the first time, the "Eddie Murphy Rule" was used; for the first time, a commodities insider-trading case reached a courtroom via a soldier and an engineer rather than a banker. The $7 billion futures layer remains open. The structural asymmetry: on-chain evidence is public; futures evidence needs subpoenas.

Pattern 8 — Rent Conversion. Crisis becomes contract: the ex-militant surveillance network (Chapter 23), the wartime logistics mandates (Chapter 27), the insurance backstops. In each case a shock converts into a durable revenue stream allocated by the state, often to actors with prior proximity to the problem — or to the cause.

Pattern 9 — The Lobbying Dual-Track. Foreign principals spend openly through registered agents (Saudi Arabia over $74 million in 2025; the UAE over $150 million since 2016 through two dozen firms, alongside documented campaign contributions by the same firms — legal, disclosed, and on record) while domestic industry spending runs through trade associations that an Exxon lobbyist, caught on camera, described as the industry's

"whipping boy" — the layer designed to absorb public blame. The pattern is not illegality; it is architecture. The same lobbying firm appeared in this document twice: registered for Riyadh, and opening Caracas.

Pattern 10 — Opacity as Strategy. Dated Brent ran $25 above futures in the crisis because assessments are human judgments in illiquid markets; Russia classified its refinery data mid-war; China has never published its reserve figure since 2017; OPEC's demand forecast functions as its production policy's input. In each case, information itself is an instrument, and the honest reader treats every quoted number as someone's output.

Pattern 11 — The Hub Squeeze Repeats. Cushing 2008 (squeeze), Cushing 2020 (negative, too full), Cushing 2026 (premium, too empty): the same delivery machine fails in opposite directions, each failure transmitted to a global price because the contract is physical.

Pattern 12 — Windfall Meets Enforcement. The windfall chapters (refiners +150 percent; traders' $250 billion supercycle of 2022–23) overlap the enforcement chapters (Glencore's record marketing profits during conduct years; Vitol's $41 billion run followed by its Venezuela mandate). The pattern is not hypocrisy — it is sequencing: penalty, monitor, early termination, license.

Pattern 13 — The Subsidy Loop. Cheap fuel breeds arbitrage; arbitrage breeds theft; theft breeds security contracts; removal of the subsidy collapses the arbitrage but ignites politics (Occupy Nigeria 2012, the gasolinazo 2017, Indonesia's perpetual balancing, Nigeria's 2023 shock that halved measured theft within two years). The loop's exit is pricing reform plus security —

and the political bill lands on whoever pulls the lever.

Pattern 14 — The First-Use Expansion. Every crisis extends the legal frontier's first use: first Eddie Murphy Rule prosecution, first extraterritorial venue for platform-data theft, first shadow-fleet criminal case, first OFAC look-through guidance, first IEEPA expansion into digital assets. The law does not anticipate the gray market; it gets drafted by it.

Pattern 15 — Forecast Chasing. The EIA's January 2026 forecast for the year's average Brent was $51; the war made it roughly $95 — a $44 error caused entirely by an unforecastable event, in the same year that OPEC's demand outlook and the IEA's diverged by the largest margin on record (over 3 million barrels a day). Every institution's number is an output of its mission. Chapter 36 asks what forecasting is even for.

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PART X — THE POLITICS OF THE NUMBER

Chapter 29: Pain at the Pump, Pain at the Polls — and the Blame Machine

Of all economic statistics, none moves politics like the number on the gas station sign — and the research record on *why* is one of behavioral economics' neater findings.

The numbers, from peer-reviewed sources: across four decades of presidential approval data, each ten-cent increase in gasoline prices shaved more than half a percentage point off approval — an effect *independent of media coverage*, unlike unemployment or inflation, because the sign is its own headline, self-published at every street corner, consulted weekly by nearly every adult. Nonlinear effects appear in more recent work: the *first* spikes of an episode do the damage; voters habituate to high levels. The mechanism seems to be pocketbook rather than sociotropic — people feel it directly, in the ritual of the fill-up. County-level analysis found block groups with heavy fuel consumption swung measurably against the incumbent; a celebrated correlation attributed to a president himself — "my approval is almost perfectly correlated with the price of gasoline" — captures what the data show.

And yet the control test is brutal: statistical analysis finds *no significant relationship* between the president's party and long-run inflation-adjusted gas prices across seven decades — for the obvious structural reason that this document has now spent thirty chapters establishing: presidents control taxes, releases,

waivers, and speeches; the crude layer belongs to geology, OPEC, traders, refiners, insurers, and wars. The asymmetry is the point: **the number is politically owned by whoever holds office regardless of who sets it** — which is why every crisis produces the same choreography of letters, investigations, windfall-tax proposals, and excise holidays documented across 2022 and again across 2026, and why, in September 2026, with the national average at $4.48, a president could tie the timing of price relief to the election calendar on camera, and why his opponents' polling blamed him. Both reactions are the political system working exactly as the research predicts — visibility economics running ahead of control economics.

Chapter 30: Windfall — The Profits Question, With the Numbers On the Table

The 2026 profit record is not in dispute; what to make of it is. Here it is, all of it, from SEC filings and company reports.

The five largest international oil companies plus the five largest American refiners earned more than $65 billion in the second quarter of 2026 alone. Marathon Petroleum's refining margin was $36.33 a barrel against $17.58 a year earlier; its net income quadrupled to $5.1 billion. Phillips 66's margin doubled. Valero earned $3.7 billion. The three majors combined returned $6.3 billion to shareholders in the quarter. Marathon's and Valero's stock prices each rose more than 150 percent in nine months. The international oil companies' tally — ExxonMobil's underlying $8.8 billion, Aramco's estimated $31.9 billion, the supermajors' combined run-rate back toward the $219 billion record of 2022 — tells the same story at the upstream layer. Meanwhile the national average pump price rose 43 percent year-over-year and diesel set an all-time record of $6.53.

The 2022 playbook repeated almost shot-for-shot: record profits, a presidential accusation of war profiteering, congressional windfall-tax bills, and — where politics differed — actual levies: Britain's Energy Profits Levy, the EU's windfall tax, and in July 2026 Portugal's 33 percent levy on oil and refining profits to fund household fuel relief. France summoned TotalEnergies' CEO to parliament. The counter-record belongs in the same paragraph: refining-margin windfalls partly reflect inventory and hedging accounting; the margin lives at the crack

spread rather than the pump; windfall taxes, economists note, deter the very investment that relieves bottlenecks (the refineries that would ease the next crisis are the ones such taxes make uneconomic to build); and the boundary between lawful windfall and unlawful conduct is defined by the enforcement docket of Chapter 22, not by the political calendar. The document holds both: the arithmetic of who gained, and the unresolved argument about who should.

Chapter 31: The Influence Layer — Following the (Legal) Money

The final layer of the politics is the least visible and the most documented: the organized effort to shape the rules under which everything above operates.

The foreign track. Under the Foreign Agents Registration Act (FARA), Saudi Arabia's government spending on U.S. influence exceeded $74 million in 2025; the UAE has spent over $150 million since 2016 through two dozen firms, conducting more than ten thousand recorded political activities — contacts with two hundred-plus congressional offices, funding for eighteen think tanks, and thousands of media placements, per a comprehensive independent study. The named architecture is on record: Akin Gump at $13.5 million (whose agents' own campaign contributions, disclosed separately, totaled over $1 million, including $528,000 to 104 members of Congress its agents had contacted on UAE business), Brunswick, Camstoll, Teneo; think-tank funding measured in the millions for the Atlantic Council and the Middle East Institute. A Trump-linked firm, Ballard Partners, registered at $60,000 a month for the Saudi embassy — and, as Chapter 27 noted, opened a Caracas office in July 2026. All legal; all disclosed; all on record.

The domestic track. Domestic oil and gas political spending — lobbying, campaign, super PAC — measured in the hundreds of millions per cycle. A Senate committee minority report in 2026 tallied $201 million in fossil-fuel-linked contributions to the 2024 cycle's winners, a $1 billion request at a 2024 dinner, a $40

million Houston fundraiser, $19 million for an inaugural ($2 million from Chevron alone), and super-PAC flows exceeding $400 million in 2023–24 — more than six times the healthcare industry's — framing the 2025 tax act's roughly $190 billion in industry benefits as the return on investment. The report is a partisan document and is labeled as such; the underlying FEC data are public; the industry counters that political participation is ordinary and the policy returns were convictions, not payments. The reader weighs.

The California laboratory. At state level, the lobbying record is itemized: oil interests spent over $17 million in California in the first half of 2026 alone — Chevron $2.55 million and the Western States Petroleum Association $2.28 million in a single quarter — against a stack of bills on refinery transparency, closure staffing, displaced-worker funds, and climate lawsuits, including the single most-lobbied bill of the session. Front groups ("Californians for Energy Independence") absorbed seven-figure company funding. Whatever one's policy view, the anatomy is the same at every scale: **the rules of the price are contested, professionally, continuously, and legally — and the pump price is where the contest's results become visible.**

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PART XI — THE WORLD MAP: WHO EATS THE SHOCK

Chapter 32: Six Ways to Smooth a Price — And One Way to Fail

When the 2026 shock hit, the world's governments ran a live, uncoordinated experiment in price management. Six designs, six results.

China: the bandwidth rule. Beijing's price mechanism resets retail ceilings every ten working days against a basket of world crudes — but adjusts only if the product price moves more than a threshold, narrows adjustments in a high-price band, and freezes below a $40 floor and above a $130 ceiling. In March 2026 the state capped an increase at roughly half the formula's output, absorbing the difference — and, on the way down in summer, passed through the largest cuts in years. Between the two, state refiners ate the margin. The design's feature: the rule is automatic, so no politician owns any single price change. The cost: opacity (the formula is partly unpublished) and refiner-margin volatility. In the crisis's second quarter, China's imports collapsed 32 percent — but customs-level analysis attributes over half of the fall to *storage policy*: Beijing simply stopped buying at $100 and drew its tanks instead. The largest marginal buyer in the world turned itself into a shock absorber, precisely because its rulebook let it.

India: the targeted bridge. With fuel formally deregulated since 2014, New Delhi's 2026 answer to the aviation-fuel crisis

was a ₹10,000 crore (about $1.1 billion) *recoverable* Price Stabilization Fund: interest-free advances to state refiners so domestic jet fuel could be capped at half its import-parity price, with the advances recouped when prices fall. A revolving bridge, not a subsidy — sharing the risk rather than eating it.

Vietnam and Thailand: the revolving fund, two ways. Vietnam drew its stabilization fund through the spring, then — in a move this document flags as significant — ended uniform pricing in October 2026, letting stations set their own prices under a formula with the fund as a backstop: deregulation *with* a safety net. Thailand ran the most sophisticated version: cap the pump price briefly, then let it rise on a schedule while clawing back "excess refining margins" from refiners during windfalls — diesel rose 36 percent against Malaysia's 80 and Indonesia's 77 over the crisis window. The fund pays for smoothing by recapturing the middleman's crisis rent.

Indonesia and Malaysia: the frozen price, the open wallet. Jakarta froze subsidized prices by decree through end-2026 (a ~$12 billion budget line, bridged by nickel and coal windfall taxes) and raised premium-tier prices to stop leakage to the rich; Kuala Lumpur's subsidy bill doubled past its budget as the state simply paid the difference. Both work; both are regressive (the IMF's own verdict on blanket fuel subsidies: the benefits flow to the heaviest consumers — businesses and the affluent); both face the same wall when prices stay high for a year.

Australia and Canada: the excise lever. Australia halved its fuel excise for three months ($2.9 billion package) alongside a

strategic-package build-out; Canada suspended its federal excise entirely for the summer. Fast, visible, expensive, and — as Chapter 14's arithmetic showed — an order of magnitude smaller than the crude move. The honest label: relief for the bill, not control of the price.

Nigeria and the Philippines: the cautionary files. Nigeria removed its subsidy in 2023 — the shock tripled pump prices, ignited protests, yet *halved measured theft within two years* and repaired the treasury; by 2026 claims persisted that billions in "subsidy-like" spending continued through the back door, and the House opened its underpricing inquiry (Chapter 23). The Philippines, where a 1984 stabilization fund — the Oil Price Stabilization Fund, OPSF — accumulated a deficit that helped force the era's deregulation, saw 2026's hundred-peso liter revive calls to repeal deregulation entirely — a full-circle debate. The design variable that separates the working funds from the failed one, across every file in the archive: **does the rule force timely pass-through, or does a politician control the valve?** OPSF failed because politicians controlled the valve. China, Thailand, and Vietnam's funds work — relatively — because the rule is automatic.

And the one way to fail: freeze the price while buying at world prices and printing the difference — Argentina's eternal cycle, Venezuela's ruin, the 1970s American allocation system of Chapter 4. Every file in the archive agrees with the one verdict economics has repeated for fifty years: suppressing the signal does not suppress the scarcity; it re-rations by queue, favor, and black market — and hands the rent to whoever controls the gap.

And Europe's entry in the 2026 experiment was different in kind: not smoothing but *suspension* — Germany cutting its energy tax from October, Spain cutting fuel taxes, Britain postponing scheduled increases, France refusing cuts as a matter of principle. The European position previews the coming layer: the ETS2 carbon market arrives on European road fuel in 2028 (the wartime oil-price contingency having already triggered its postponement once), adding roughly fifty to seventy cents a gallon to the European stack as a deliberate, itemized, policy choice, cushioned by the Social Climate Fund — the same architecture California already runs. The world map of pump prices is becoming a map of climate policy, line item by line item.

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PART XII — SHOCK HISTORY: A CENTURY AND A HALF OF PRICE

Chapter 33: Eleven Shocks — The Chronology That Explains the Present

The price of gasoline has a memory, and every crisis teaches the market a lesson it never forgets. Here is the full chronology, with the lesson each shock left behind.

1860–1861: the original crash. The Drake well (Titusville, Pennsylvania, August 27, 1859) launched an industry by proving oil could be drilled rather than skimmed from seeps. Within a year, production quadrupled and the price of oil fell from $20 a barrel to ten cents — "sometimes cost less than water." The first boom-bust taught the first lesson: *oil's natural state is oversupply*, and every stabilization regime since — Standard Oil, the Texas Railroad Commission, OPEC — has been a response to that chaos. The Civil War's alcohol taxes (raising liquor to $2 a gallon against a dime on petroleum) produced history's first oil-price spike, $10–13 a barrel: the first lesson that *tax architecture is price architecture*.

1870–1911: the monopoly stabilizer. Standard Oil's 90 percent control stabilized prices as intended — and was broken by the Supreme Court for the methods used to achieve it, seeding every modern oil major (Chapter 3).

1930–1931: capture kills the market. The East Texas flood drove local crude to ten cents a barrel; martial law and the Railroad Commission's proration invented the swing producer

(Chapter 3). The inflation-adjusted pump-price records of 1918 and 1934 ($4.88–$5.71 in today's dollars) stand as monuments to *policy-created* scarcity: Prohibition's fuel-alcohol taxes and Depression-era production quotas.

1973–1974: the embargo. OAPEC cut production and embargoed the West; crude quadrupled to nearly $12 in three months. The visible lesson — dependence is leverage — drove the IEA's creation, America's strategic reserve, and the CAFE efficiency regime. The invisible lesson of Chapter 4: price controls turned the shock into gas lines.

1979–1980: the second shock. Iran's revolution removed 4.5 million barrels a day; prices doubled again; the inflation-adjusted U.S. pump price hit $5.08 (1981) — a record that stood for three decades.

1986: the collapse. Saudi Arabia abandoned price defense for market share; crude fell by half in weeks. The lesson that would govern the next three decades: *the swing producer sets the ceiling, and the swing producer is whoever has the spare capacity and the discipline.*

1990–1991: the Gulf War spike — brief, because Saudi spare capacity filled the gap; the first strategic-reserve release.

2003–2008: the super-cycle. China's industrialization lifted demand against a stagnant supply base; crude ran to $147 in July 2008 and American gasoline to $4.11 — the nominal record until 2022. The crash that followed (two-thirds in five months) introduced the modern era of correlated risk: oil had become an asset class.

2005: the hurricane test. Katrina and Rita knocked out roughly a quarter of America's refining capacity in a single month, and the national pump price jumped 46 cents in a week — the largest weekly spike in the modern data. Washington's response set the template for every later gouging debate: the Federal Trade Commission conducted an exhaustive investigation and found *no evidence* that refiners had manipulated prices, withheld capacity, or manipulated inventories — price increases were, in its words, approximately what a competitive market would predict. Fifteen of seventy-seven examined firms met the statutory definition of "gouging," but regional market trends explained nearly all cases. The honest reading, valid ever since: storms break regional supply, prices spike locally, and the spike is the market rationing scarcity — the gouging question turns on conduct after the spike, not on the spike itself. A decade later, Hurricane Harvey supplied the counter-illustration this document uses elsewhere: with a third of Gulf Coast refining down, *crude prices fell while gasoline soared* — the canonical portrait of the crack spread (Chapter 9) doing exactly what it is.

2014–2016: the share war. OPEC's Thanksgiving 2014 decision not to cut — a deliberate choice to defend market share against American shale — cratered prices from $115 to under $30 by January 2016, and taught the shale industry the capital-discipline religion it still preaches: *grow at any price died in that crash*.

2020: the negative barrel. COVID erased a sixth of world demand; the Russia–Saudi price war filled the tanks; WTI went below zero at Cushing (Chapter 7). Refineries worldwide closed;

the closures became 2026's bottleneck (Chapter 10).

2021–2022: the invasion shock. Russia's war on Ukraine removed 2–3 million barrels a day of supply and risked far more; American gasoline hit its all-time nominal record of $5.01 (June 14, 2022); the strategic reserve was drawn by 180 million barrels; the G7 invented the price-cap regime; the industry earned its $219 billion year. Diesel — the forgotten fuel of the economy — spiked hardest, teaching the market to watch *products*, not just crude.

2026: the closure. And then the largest supply shock in the history of the oil market — the subject of the next part.

Through all of it, the real-price ceiling barely moved: adjusting for inflation, the most expensive gasoline Americans have ever bought was not June 2022 but *July 2008 and 2012*, at over $5 a gallon in today's money — and the all-time inflation-adjusted annual peak remains 1918, a wartime artifact. The sign at the station is dramatic; the long chart is remarkably flat. Prices rage; the real ceiling endures.

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PART XIII — THE 2026 WAR: ANATOMY OF THE LARGEST SHOCK ON RECORD

Chapter 34: The Year Oil Went to War

The story of 2026 is the story of what happens when every lesson of Chapter 33 is tested at once: a chokepoint, a refining bottleneck, an insurance market, a sanctions regime, a demand structure with no slack, and a war — converging on a single strait.

The origin. The long arc ran from America's 2018 withdrawal from the Iran nuclear deal through Iran's steady enrichment toward weapons-capability thresholds, the June 2025 "Twelve-Day War" in which Israel and the United States struck Iranian nuclear sites, a failed diplomatic track through late 2025, and then — on February 28, 2026 — American and Israeli strikes on Iran. The stated rationales recorded in the archive span the full range: preemption against a nuclear threshold capability, response to Iranian regional attacks, support for domestic protesters, and (claimed by critics and never proven) resource motives. This document records all of them, adopts none, and notes the institutional pattern: causation in war is the most contested claim in any archive.

The closure. Within days the Strait of Hormuz — through which a fifth of the world's oil and a fifth of its liquefied natural gas normally pass — was closed, not by mine-clearing navies but by the mechanism of Chapter 12: insurance. Tanker traffic collapsed from over a hundred transits a day to single digits and, at the worst, one. Crude exports from the Gulf fell by roughly

three-quarters; Iraq's exports fell nearly 90 percent in a single month; the net loss to world supply — about fourteen million barrels a day, after Saudi and Emirati bypass pipelines ran flat-out — was *fourteen times* the Ukraine war's disruption and triple the 1973 embargo's. Brent ran to $118–130; the physical "dated" price briefly beyond that.

The machine's response — and its limits. The system's stress responses, each documented: the largest strategic release in history (400 million barrels across 32 countries, about 80 percent of it delivered by October, at roughly 2.5–3.3 million barrels a day — weeks of cover, not months); the Jones Act waiver (real, small); the idle reinsurance facility and the escorts that never came (Chapter 12); China's storage drawdown (imports cut by a third, half of it policy); refinery utilization at 98 percent, and then the autumn maintenance season. And on the demand side, the system did what demand does at these prices: global consumption fell by roughly 2.5 million barrels a day in 2026, with quarterly drops of five million or more at the worst — the deepest non-COVID demand destruction ever recorded, concentrated in petrochemicals and developing Asia.

The product squeeze. With crude fungible and refineries fixed, the war's sharpest edge was products (Chapter 10): European crack spreads at all-time records; American gasoline cracks above $40; jet fuel cracks at $80; diesel's national average at $6.53 — the highest price ever posted on an American pump sign. The countries that moved fastest on the product side — the G7's second release round, diesel-first; the EU's emergency diesel-stock discussions; Australia's strategic package — were

responding to the correct diagnosis: the world was not out of oil; it was out of *deliverable fuel*.

The geography of pain. America, insulated by its own production and refining, ran $4.40 gasoline — painful, but a third below Europe's crisis peaks and far below Asia's subsidy-burdened economies. California, with its island refining market, touched past $6.40 — and its September emergency switch to winter gasoline under its new standby authority was itself a small lesson in how much of the state's premium is plumbing rather than policy. Subsidized Asia bore the fiscal cost; Nigeria and the Philippines lived the political one.

The exits. The war's negotiating history was itself an oil-market instrument: a Pakistan-brokered ceasefire in April, a framework memorandum of understanding in June accompanied by the most expansive Iran oil license ever issued — revoked within fifteen days after renewed attacks — a Versailles summit in August, a second memorandum, and by October a standoff in which Iran proposed a reopening plan, Washington publicly rejected it, Brent sat above $100, and roughly one ship a day transited a strait built for eighty-five. As this document closes, the base-case forecast held by most institutions is normalization through 2027 with Brent in the $70s — and the minority view, worth respecting after a year in which every consensus was wrong, is that the corridor war has reset the risk premium permanently.

The standings, October 2026. Brent near $100–102, backwardated $7+. American regular gasoline $4.37 and falling about eleven cents a week as refineries exit maintenance; diesel

at its all-time record; the strategic reserve at 285 million barrels, lowest since 1983, with a refill bid building underneath the market; OPEC's core producers running five million barrels a day below pre-war levels; and a global inventory position that, by year-end, will stand about fifty days of demand — the thinnest cushion since modern records began. Whatever comes next in the diplomacy, the physics of 2026 has already rewritten the economics: spare capacity, bypass pipelines, storage, insurance, and refining redundancy are now priced into every barrel on Earth.

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PART XIV — DEMAND: FIVE REGIMES, ONE CURVE NO MORE

Chapter 35: Who Is Still Buying — The Geography of Thirst

"World oil demand" is a fiction — a sum of five different behaviors that respond to five different forces. Understanding the pump price's future means understanding that the five have decoupled.

The United States: the inelastic giant. America consumes more gasoline than the next dozen economies combined — about 8.9 million barrels a day in 2025 — yet its *per-capita* consumption sits at 1967 levels, because fleet efficiency has quietly eaten three decades of growth. The demand curve here is startlingly price-blunt: through five months of $4.40-plus gasoline in 2026, weekly consumption fell only 1–2.5 percent — the elasticity evidence of Chapter 6 confirmed in real time. A record 72 million Americans drove over the July 4th weekend anyway. Demand destruction here is real but slow, driven by the vehicle stock's decade-long turnover, not by price.

Europe: the managed decline. European oil demand peaked in 2022 and is falling about 120,000 barrels a day annually — diesel first, as the dieselization era reverses itself. New-car sales are now one-third gasoline, one-tenth diesel, and over half electrified in some form. Europe's demand is policy-driven, price-cushioned by taxes (so crude swings barely register at the pump), and structurally terminal. In 2026 its refined-product demand fell further even as refinery margins hit records — the

decline and the windfall in the same year, in the same market.

China: the structural break. The most important demand story in the archive. In 2026, roughly two-thirds of China's new cars were new-energy vehicles; gasoline-only model sales fell 40 percent year-over-year; electric trucks, barely a tenth of the fleet, already displace a quarter-million barrels a day of diesel because they run the heavy-utilization routes where electrification pays fastest. China's total oil demand is now forecast to *fall* nearly 9 percent in 2026 — the steepest non-COVID decline on record — led by road fuels, with the only growth in petrochemical feedstocks. The world's largest crude importer has become a strategic accumulator: it bought a record 11.6 million barrels a day in 2025, paused completely at wartime prices, and resumed buying on the dips. Chinese demand is no longer a growth engine; it is a policy instrument.

India: the last growth engine. Against this, India still grows — gasoline demand rising 3–4 percent a year, a national refining build-out racing to match it, the government capping fuel prices with recoverable funds when shocks hit. Every forecast of "peak oil demand" is, at bottom, an argument about when India's growth stops outrunning China's decline plus the OECD's decline. The forecasters disagree on the date by a decade; they agree on the arithmetic.

The developing subsidy belt. Indonesia, Malaysia, Egypt, Nigeria's aftermath: a belt of economies where the state stands between the world price and the pump, and where demand responds to politics, not price. Their 2026 story was fiscal: budgets doubling, windfall taxes invented to pay for frozen

pumps, targeted rationing by income decile.

The elasticity verdict. The academic literature the archive assembled spans an order of magnitude: aggregate time-series studies find American short-run gasoline elasticity near zero; tax-shock and micro-data studies find it three to ten times larger. The reconciliation: the *aggregate* curve is blunt (people must drive), the *marginal* behavior is elastic (they merge trips, defer purchases, switch modes at the edges), and the *long run* is very elastic indeed (they change cars, homes, and cities). The practical consequence, priced into every shock in this document: **in the short run, almost the entire burden of a supply shock lands on price; in the long run, almost all of it lands on demand.** The demand curve is the shock absorber — it just engages slowly, and the lag is where the crises live.

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PART XV — BEYOND THE PUMP

Chapter 36: The Rest of the Barrel — Jet, Ships, Plastics, and the Station of the Future

Gasoline is barely half the story of a barrel of oil. The other half is where the next decade's price battles will be fought.

The skies. Aviation fuel — roughly 8 percent of oil demand — spent 2026 in the most extreme margin environment ever recorded: jet cracks that historically stayed under $20 touched $80; airlines, whose fuel is 30–60 percent of operating cost, raised fares into 2027 to rebuild margins. And the policy layer is arriving: sustainable-aviation-fuel mandates (2 percent in Europe in 2025, rising to 70 percent by 2050; UK and Korea joining) at two-to-five times conventional jet cost, the EU's aviation carbon exemption ending in 2026, and — a crisis irony — high oil prices compressing the green premium exactly when mandates require buying. Airport fuel pricing is becoming as policy-laden as the European road-fuel stack.

The seas. Shipping's 2020 sulfur-cap switch (IMO 2020) was the dress rehearsal for product-market convulsions: three million barrels a day of marine fuel had to change recipe, scrubber economics boomed and busted with the spreads, and the pattern — a regulatory spec change rippling through crack spreads — previews the coming greenhouse-gas framework, which spent 2026 adjourned and contested between the nations that want a levy and the ones that call it an attack on their fleets.

The molecule economy. The most consequential demand fact of the decade: petrochemicals — the plastics, fertilizers, and synthetic materials made from oil and gas — are now *more than 95 percent of net global oil-demand growth*, approaching one-sixth of all consumption by 2030. The world's refineries have noticed: India's Jamnagar, China's new mega-complexes, Saudi Arabia's downstream empire are all configured to chase molecules, not miles. The strategic consequence for the reader: even a world of perfect electrification retains an oil demand floor — the argument over whose size splits every forecaster in this document.

The station. And what of the pump itself? The American station — 148,000 strong, convenience-store-cored — faces its long transition. The amenity-rich giants (Sheetz, Wawa, Buc-ee's) are converting forecourts into charging-and-food destinations, betting that twenty-three-minute charging dwells can be monetized like the c-store; the small urban and rural stations face the triple squeeze of falling gallons, underground-tank liabilities, and land values. The forecasts range from a gradual consolidation to a claim — from a major consultancy — that four in five conventional stations could be unprofitable by 2035. Against that, the counter-facts: America's record vehicle miles traveled, gasoline demand down barely 1.5 percent in the worst price year since 2008, and an EV fleet still in single digits. The pump's obituary, like oil's, has been written prematurely before.

The niches. The archive's smallest files carry their own lessons: avgas — the last leaded fuel on Earth, now mid-transition to unleaded under a 2030 mandate, and priced above premium

car gasoline precisely because lead was the cheapest octane ever found; propane, the parallel heating market with its own freight layer; dyed diesel, the tax-exempt twin whose spread with road diesel is, down to the cent, a map of fiscal architecture. Every fuel price is a policy document written in molecules.

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PART XVI — FORECASTING, AND THE FUTURE OF THE NUMBER

Chapter 37: The Honest Impossibility of Knowing — And What To Watch Instead

In January 2026, the U.S. Energy Information Administration forecast that Brent crude would average $51 that year. The war made it about $95 — a $44 error, and the entire error was one event nobody forecast. The same year's most careful academic work showed that raw futures prices are barely better than a coin flip a year out (though careful combinations of futures and models do modestly better), and the entire forecasting industry flipped from predicting a 2026 glut to predicting a 2027 one within a single summer. This document therefore declines to predict prices — and instead ends with the variables that decide them, which is more useful.

Watch spare capacity and its owners. The swing producer — Texas 1930s, OPEC 1980s, Saudi Arabia-plus-whoever 2020s — sets the ceiling (Chapter 3). The 2026 drama added a new variable: an unconstrained UAE racing toward five million barrels a day outside the cartel, and an Iraq tempted to follow. If OPEC's quota system fractures, the ceiling mechanism dies with it and prices discover their floor instead.

Watch the middle of the chain. Refining is the new bottleneck (Chapter 10): capacity additions through 2030 are a fraction of what a resilient system needs; every war and every drone now targets the same node. Crack spreads — the refining

margin — are the single best early-warning indicator of pump trouble, ahead of crude itself.

Watch insurance and chokepoints. The 2026 lesson is that *insurable flow* is the true supply (Chapter 12). War-risk premia, P&I coverage decisions, and the construction of bypass pipelines are leading indicators of the next risk premium.

Watch the five demand regimes separately (Chapter 35): China's storage behavior, India's growth, America's vehicle fleet, Europe's decline, the subsidy belt's fiscal stamina. The global "demand" number is a sum that hides the moving parts.

Watch the restock bid. America's refill, Japan's replacement barrels, China's buying, India's build — strategic tank-filling is a floor of price-insensitive demand under 2027 that most supply-glut forecasts ignore.

Watch the legal frontier (Part VIII): the December 2026 war-trades trial, the SRE dockets, the shadow-fleet prosecutions, the prediction-market legislation — each will redraw some boundary between lawful and unlawful information in the price.

Watch the policy stack. Carbon pricing arrives on European road fuel in 2028; America's highway fund faces its 2028 insolvency; California's experiment continues to export its results — the tax stack (Chapter 15) is the most predictable component of every future pump price, because it is chosen, not discovered.

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APPENDIX I — THE MASTER TIMELINE (selected anchors)

1859 Drake well, Titusville — oil $20/bbl → 10¢ within two years. 1870 Standard Oil founded. 1882 the Trust. 1905 first drive-in station, St. Louis. 1911 Supreme Court dissolution. 1921 tetraethyl lead discovered. 1930–31 East Texas flood; martial law; prorationing. 1938 Mexico expropriates. 1945–73 posted-price stability, $1.80–$2.50/bbl. 1960 OPEC founded. 1970 US production peaks; Clean Air Act. 1973 embargo; $11.65/bbl; IEA and SPR follow. 1975 SPR authorized; last US refinery wave. 1977 OPIS founded. 1979–80 second shock. 1982: 301 US refineries. 1983 WTI futures launch. 1984 Philippines OPSF. 1986 OPEC collapse; Kharg Island battered. 1989 Exxon Valdez → OPA 90. 1990 CAAA boutique-fuel era. 1991 "rockets and feathers" coined. 1993 federal gas tax frozen at 18.4¢. 1997 Mexico gasolinazo. 1998 Philippines deregulates; OPSF abolished. 2005 Katrina; FTC clears industry. 2008 $147/bbl; $5.10 real pump record; Cushing squeeze cases. 2011–14 shale boom; WTI discount era. 2014–16 OPEC share war. 2015 export ban lifted. 2016 US gasoline record year (143.2B gal). 2020 negative WTI (−$37.63, Apr 20). 2021 Colonial ransomware. 2022 Ukraine invasion; $5.01 record; $219B profits; 180M-bbl SPR drawdown; G7 price cap. 2023 Nigeria subsidy ends; CMA fuel study. 2024 UAE capacity 4.85M b/d; TMX opens; Grangemouth closes. 2025 third straight year of falling US pump prices ($3.10 avg); China record imports; E15 waiver; Enverus $70 shale breakeven. 2026: Feb 28 war; Hormuz closes; Brent $118–130; 400M-bbl IEA release; May 1 UAE leaves

OPEC; June 14–15 MOU; July re-escalation; Sept Petroline struck; Oct 2 G7 round two. Standings at writing: Brent ~$102, US regular $4.37, diesel record $6.53, SPR 285M bbl, Hormuz at ~1 transit/day.

APPENDIX II — LAWS BROKEN AND LAWS MADE (the index)

FCPA (foreign bribery of officials) — Vitol, Gunvor, Trafigura, Glencore, Freepoint resolutions. Travel Act and money-laundering statutes (18 U.S.C. 1956/1957) — parallel counts across the docket. Commodity Exchange Act §§6(c), 9(a), Rule 180.1 — manipulation and anti-fraud: Optiver, Parnon/Arcadia, Glencore, Shak, Trafigura, TotalEnergies settlements; DRW and EOX losses define the boundary. The "Eddie Murphy Rule" (Dodd-Frank §746; CEA 4c(a)) — first used 2026 (Van Dyke). IEEPA and sanctions executive orders (13224, 13846, 13902, 14024) — the shadow-fleet designations. UK Bribery Act — Glencore UK; SFO's Unaoil quashals. Sherman Act 1911 (Standard Oil); Robinson-Patman (gas wars); EPAA 1973 (the control era); CA SBX1-2 (gouging statute); EU ETS Directive 2023/959 (carbon on road fuel, 2028). RFS/Clean Air Act layers (the RIN apparatus; RVP; boutique blends). Never successfully applied: war-profiteering statutes to commodity markets — the legal gap at the heart of the windfall debate.

APPENDIX III — SOURCES, PEDIGREE, AND HONEST LIMITS

This document synthesizes sixteen research dives compiled between October 5–6, 2026, comprising several hundred pedigreed sources graded by authority: official statistics (EIA, IEA, OPEC, customs, regulators, courts) at the top; major financial media next; industry and advocacy material always labeled as such; analyst estimates labeled as estimates. Conflicting evidence is presented in place, not averaged away. Specific honest limits the reader should hold: several figures remain single-sourced (flagged where they occur); the 2026 war was *ongoing* at this writing and its later chapters will continue to move; the prediction-market and futures investigations were open, with charges announced against two individuals and none against the institutional traders; allegations against public figures in Nigeria, Mexico, Brazil, and elsewhere are recorded with their current procedural status — unresolved, dismissed, or annulled — and no unresolved allegation in this document should be read as a finding.

APPENDIX IV — WHAT THIS DOCUMENT DID NOT DO

It did not assign blame. It assembled the map — geology to molecules, molecules to money, money to law, law to politics — and placed on it every documented flow: the lawful, the unlawful, the ambiguous, and the still-open. The pump price, it turns out, is not a number set by a villain or a market. It is a sedimentary record: of ancient seas and modern wars, of refiners and teapots, of treaties and trials, of things chosen and things imposed. You now hold the strata. What they mean — at the ballot box, the policy hearing, the investment committee, the dinner table — is yours.

The sign at the corner will change tomorrow. Now you know everything that moves it.